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Grid Impact Assessment and Energy Optimization of the new Electrified Port of Oskarshamn

Smart Charging, Onshore Power Supply
and Local Flexibility Resources

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**Grid Impact Assessment and
Energy Optimization of the
new Electrified Port of
Oskarshamn**

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Abstract

Port electrification reduces Greenhouse Gases (GHG) and other particulate emissions but can create high, variable loads that stress distribution networks. This thesis assesses the planned electrification of the Ferry and Cargo Terminals at the Port of Oskarshamn (Sweden), including cargo-handling Electric Vehicle (EV), Onshore Power Supply (OPS), Photovoltaic (PV) and Battery Energy Storage System (BESS), evaluating how flexibility can reduce grid impacts and electricity costs.

A four-stage lexicographic Mixed Integer Linear Programming (MILP) charging optimization model was implemented in Pyomo and solved with Gurobi. Time-series power-flow simulations in pandapower then compared conventional charging, optimized charging and PV-BESS scenarios.

Optimized charging reduced annual electricity costs and peak demand at both terminals, while substantially limiting the impact of EV charging on the grid. At the Ferry Terminal, smart charging combined with PV and BESS reduced critical line loading, eliminated transformer overloads and stabilized bus voltages. However, connecting both terminals to the same feeder caused severe and prolonged overloads, confirming the need for a separate grid connection for the Cargo Terminal.

Overall, the upstream feeder is the main constraint, while smart charging, PV and BESS can simultaneously reduce grid stress, reinforcement needs and Port operating costs.

Keywords

Port electrification, smart charging, Onshore Power Supply (OPS), Mixed-Integer Linear Programming (MILP), distribution grid impact, battery energy storage systems, photovoltaic generation, power-flow analysis.

Sammanfattning

Elektrifiering av hamnar minskar utsläppen av GHG och andra partikelföroreningar, men kan samtidigt skapa höga och varierande laster som belastar distributionsnäten. Denna avhandling utvärderar den planerade elektrifieringen av färje- och godsterminalerna i Oskarshamns hamn (Sverige), inklusive EV för godshantering, OPS, PV och BESS, samt analyserar hur flexibilitet kan minska påverkan på elnätet och elkostnaderna.

En lexikografisk laddningsoptimeringsmodell i fyra steg baserad på MILP implementerades i Pyomo och löstes med Gurobi. Därefter genomfördes tidsserieanalyser av effektflöden i pandapower för att jämföra konventionell laddning, optimerad laddning samt scenarier med PV och BESS.

Optimerad laddning minskade de årliga elkostnaderna och effekttopparna vid båda terminalerna, samtidigt som påverkan från laddning av EV på elnätet begränsades avsevärt.

Vid färjeterminalen minskade smart laddning i kombination med PV och BESS den kritiska belastningen på ledningarna, eliminerade överbelastning av transformatorer och stabiliserade spänningarna i nätets noder. Att ansluta båda terminalerna till samma matarledning orsakade dock allvarliga och långvariga överbelastningar, vilket bekräftar behovet av en separat nätanslutning för godsterminalen.

Sammantaget är den överliggande matarledningen den huvudsakliga begränsningen, medan smart laddning, PV och BESS samtidigt kan minska belastningen på elnätet, behovet av nätförstärkningar och hamnens driftskostnader.

Nyckelord

Hamnelektrifiering, smart laddning, landström (OPS), heltalslinjär programmering (MILP), påverkan på distributionsnätet, batterienergilagringssystem,

solcellsproduktion, effektlödesanalys.

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Acronyms

BESS	Battery Energy Storage System
CAPEX	Capital Expenditure
CHP	Combined Heat and Power
DSO	Distribution System Operator
EV	Electric Vehicle
GHG	Greenhouse Gases
ICE	Internal Combustion Engine
KPI	Key Performance Indicators
MILP	Mixed Integer Linear Programming
NPV	Net Present Value
OPS	Onshore Power Supply
PV	Photovoltaic
RE	Renewable Energy
SOC	State of Charge
SSR	Self-Sufficiency Ratio

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Chapter 1

Introduction

The global energy industry is going underneath a profound transformation: the need to mitigate climate change and to reduce CO_2 emissions and other particulates, push international climate commitments and national decarbonization strategies to accelerate the transition from fossil fuels to cleaner forms of energy. This transition is not related to new infrastructures, but also to the existing ones, which may not be completely flexible but can be retrofitted to have better performances.

Moreover, the electrification of utilities, transportation, and industrial processes is reshaping the demand profiles, leading to higher peaks that can no longer entirely satisfied by dispatchable generation technologies such as oil and gas powered plants, and are instead increasingly covered by renewable energy sources with inherently variable output.

As a result, energy networks need to supply sufficient power, but also to operate in a flexible, reliable and resilient manner: only in this way the integration of renewable energy sources, storage systems and advanced control strategies can be possible. In the global scenario, ports are responsible for roughly the 4% of global Green House Gases (GHG) emissions [1], showing particular relevance in their electrification and conversion into sustainable hubs.

1.1 Oskarshamn and its port

Oskarshamn is an industrial city in the southern coast of Sweden along the Baltic Sea, with a population of roughly 27,000 people [2]. Industries in manufacturing, energy and business services give work to more than half the city citizens, being it the engine of the region. Particularly, the energy sector is the most relevant one, with the nuclear power plant OKG, the Swedish Nuclear Fuel Management (SKB) and the innovative Center for New Energy Technology (CNETO).

Moreover, its port, one of the most heavily polluted in Sweden until 2018 [3], it is now converting and becoming an important hub for offshore wind and energy projects. Particularly, RWE, global leader in offshore wind, is collaborating with the Port of Oskarshamn, operated by Smålandshamn AB. The two entities signed a Memorandum of Understanding (MoU) to explore whether the port could be a suitable harbour for the logistics, installation, loading, operations and maintenance activities relating to RWE's planned offshore wind projects in the Baltic Sea [4].



Figure 1: Oskarshamn port [4]

The port, which manages 1.5 Mt (megatons) of goods, hosts roughly 550,000 passengers per year and has a total length of 2300 meters, is undergoing an extension process, evaluating the construction of two new terminals, one related to the operations of ships transporting goods, the other one to ferries moving passengers to Gotland. Figure 2 shows in red the new ferry terminal area, and in yellow the cargo one.



Figure 2: Port showing two new areas. Satellite image from Google Maps [5]

The terminal addressed to passengers (yellow in Figure 2) is planned to be built in five to seven years, while the one related to goods' ships (red in Figure 2) in roughly ten to fifteen years. Smålandshamn AB goal is to 100% power all the loads of these new areas with electricity, raising the need of an evaluation of the new demand, its optimization and the integration of renewable energy sources, together with the need to meet grid requirements and boundaries.

1.2 Distribution Network Overview

The current electrical grid is operated by Oskarshamn Energi AB, which is the local Distribution System Operator (DSO), owning the network and a Combined Heat and Power (CHP) biomass plant, together with a back-up one oil-powered. The biomass CHP plant has a thermal output of 17.7 megawatts (MW) serving the local district heating system and an electrical output of 3.8 MW. The CHP plant also has a flue gas condenser to provide additional 4 MW thermal output to district heating network [6]. The electrical power not provided by the plant is supplied by E.ON via a receiving station at 130 kV, which feeds two substations converting the 50 kV incoming to 10 kV. The two stations serve approximately 12,500 customers, the 54% of the ones requiring

low voltage electricity [7].

Historical peaks reached 52 MW (for seven consecutive hours maximum), against the 58 MW totally available [7]. Future development include raising to 88 MW the capacity of the local grid, building a new substation.

1.3 Stakeholders and Benefits

The three stakeholders involved are Smålandshamn AB representing the port, Oskarshamn Energi being the DSO, and finally NOVA AB, representing the municipality (Oskarshamns Kommun). The three of them have been synergistically working for years on the development of the port. Their objectives are respectively recognized in expanding and technologically improving the port, testing the robustness of the grid and receiving feedback for possible improvements, and bring technological value to the city and partnering with universities. The cooperation of the three of them is retained fundamental to the progress of this work.

The project benefits the three stakeholders by providing a detailed report of whether the current grid is able to support the port extensions and its full electrification. Moreover, by optimizing flexible load behaviours, the simulations aim to minimize grid stress, indirectly impacting on local industries and residents, thus benefiting from reduced risks of capacity bottlenecks and better planning of grid investments.

1.4 Sustainability

In this work, sustainability aspects are central: the integration of PV systems and battery storage, together with the full electrification of port operations will enable Oskarshamn to significantly reduce CO_2 emissions from fossil fuel-powered equipment, aligning with EU directives and improving the state of health of the citizens.

These measures contribute directly to SDG 7 "Affordable and Clean Energy", by promoting renewable energy integration and utilization of electric-powered vehicles, to SDG 9 "Industry, Innovation and Infrastructure", through the development of more sustainable and resilient energy infrastructure for the port and the local grid, and to SDG 13 "Climate Action", by supporting the decarbonization of port activities.

Furthermore, the reduction of local air pollutant emissions improves citizens' health and urban environmental quality, contributing to SDG 3 "Good Health and Well-being" and SDG 11 "Sustainable Cities and Communities" [8].

Relevant literature and ongoing research on this topic are further discussed in Section 2.1.

Chapter 2

Literature Review

2.1 Sustainable development of port electrification

Ports are energy-intensive infrastructures and important GHG emissions contributors, hosting several activities as berthing vessels, cargo handling equipment, terminal vehicles and auxiliary services. As EU environmental regulations become stricter, the maritime sector is aligning with decarbonization measures. In this picture, port electrification has become a central strategy to reduce GHG emissions, benefitting local air quality and noise while maintaining efficient port operations [9].

Recent literature describes ports as energy transition hubs and not only as logistic nodes, highlighting their role towards industrial decarbonization. From this perspective, maritime transport, electricity infrastructure, renewable generation and storage systems interact with industrial activities in an integrated approach in which electricity demand, local generation, storage technology and grid capacity are evaluated together [10] [11].

Port electrification can include several strategies and methods, such as OPS for berthed vessels [12], the replacement of oil-fueled heavy vehicles with electric alternatives, and the integration of Renewable Energy (RE) resources. These solutions, while reducing local emissions, represent a new technical challenge, since shore-side electricity supply and high power EV charging may create large and irregular demand peaks directly impacting on the local grid [13].

Due to this reason, port activities and impact assessments should be aligned with

real port activities, since simplified approach could fail to capture the operational complexity of port demand and operations [14]. This is the approach followed by this thesis, based on robustness and suitability of the inputs data and methodology used, representing a system aligned to referenced reality.

The increasing importance of this topic is further reflected in European policy measures, identifying European ports as climate change and energy-related infrastructure priorities. FuelEU [15] is a key directive to follow towards decarbonization, while the European Sea Ports Organisation (ESPO) 2024 Sustainability Report [16] shows how ports are accelerating investments in decarbonization measures through electrification initiatives, such as OPS, renewable energy integration, and more efficient energy management systems.

This thesis follows an integrated perspective, merging OPS and EV smart charging optimization with photovoltaic generation and battery storage, lastly analyzing the distribution grid power flow. This approach makes it possible to evaluate not only the additional energy demand created by electrification, but also its impact on the local electrical network and the potential of flexibility measures in an industrial context to reduce grid stress and operational costs.

2.2 Onshore Power Supply (OPS)

OPS, also known as shore-side electricity or power, or cold ironing, enables vessels at berth to be powered by the port electrical infrastructure, switching off their auxiliary engines. In fact, during conventional port operations, auxiliary diesel or generally oil-powered engines supply onboard electrical loads such as HVAC, lighting and other hotelling services. OPS therefore transfers this energy demand from ship-based fuel combustion systems to shore side electricity, aiming to reduce local emissions and noise in the port area, usually located nearby or inside the city area [17]. Cold ironing is considered as one of the key electrification technologies for maritime decarbonization, particularly because it directly targets emissions produced while ships are berthed, when they have the greatest impact on local communities [17]. This thesis aims to explore the technical feasibility and competitiveness of OPS implementation in the Port of Oskarshamn two new terminals, contributing to the limited number of research literature and development solutions assessed [12], evaluating its adoption and allowing the port and shipping industries to enjoy the associated benefits.

OPS is also central in a EU regulatory perspective. Under the FuelEU Maritime framework, passengers and containers ships weighted more than 5,000 GT must use OPS or alternative zero-emission technologies, from 1 January 2030 in ports covered by the Alternative Fuels Infrastructure Regulation, and from 1 January 2035 in all EU ports that develop OPS capacity [15].

From a technical perspective, OPS requires both shore-side and ship-side equipment and infrastructure. The system typically includes a grid connection point, transformers, switchgear and protection systems, frequency conversion when required, cable management systems and connection equipment on the shore-side. On the vessel side, ships must be equipped with compatible electrical interfaces and onboard systems able to receive power from shore [18]. The ISO IEC/IEEE 80005-1 standard defines requirements for high-voltage shore connection systems used to supply ships with electrical power from shore, with technical standardization being essential to ensure safe and interoperable operation between vessels and ports [19].

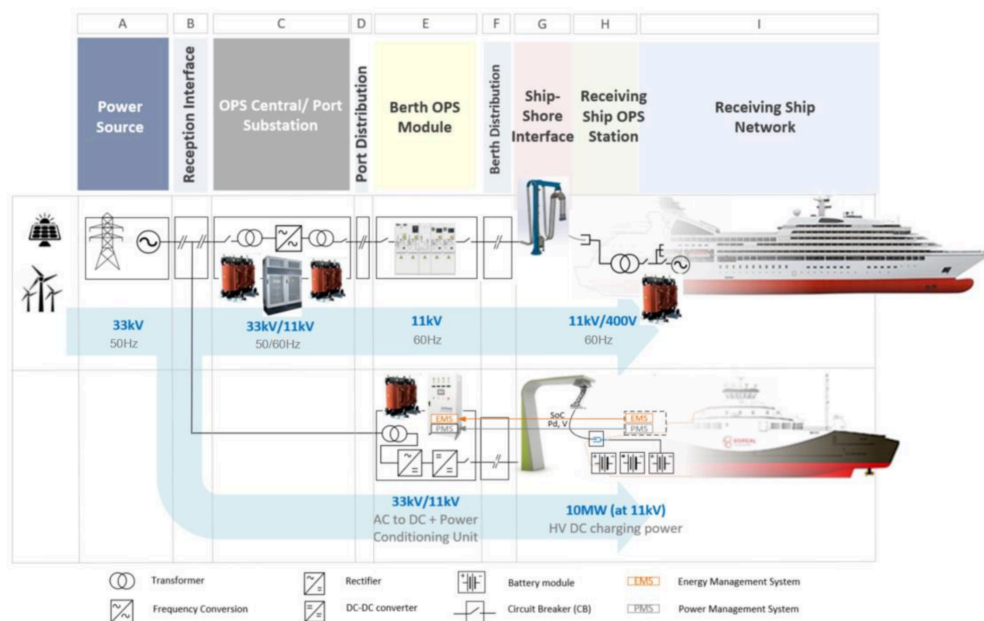


Figure 3: Key infrastructure elements for OPS [18]

Despite some of its already cited benefits, OPS development faces several barriers. In fact, the uptake of this technology has historically been limited by mainly four obstacles: technology and operations, institutional elements, economic elements and stakeholder elements. Particularly, the high investments costs, coupled with uncertain

utilization rates and lack of standardization among ships have affected OPS projects feasibility. These barriers co-exist with electricity pricing and responsibility between port authorities and shipowners, while grid connection limitation require also DSO collaboration and investments [20].

A critical and focal problematic for OPS and this work is this technology impact on the local electrical grid. While OPS reduces emissions at the berth, it introduces a large and irregular energy demand, with rapid peaks mainly depending on the vessel type, installed auxiliary engine power and duration of the stay. Consequently, OPS increases line loading, transformer loading and voltage deviations, with its planning that should not only consider the port-side connection infrastructure but also the capacity and operating limits of the distribution grid [17].

2.3 Cargo-handling equipment electrification

Cargo-handling equipment is one of the main impacting components of port operations energy consumptions. It comprehends forklifts, terminal movers, reach stacker and loaders, all conventionally powered by Internal Combustion Engine (ICE) based on diesel. Due to their energy intensive operations, cargo handling equipment is often considered as one of the most significant sources of air pollution caused by terminal procedures [21].

Terminal tractors, forklifts, reach stackers and other cargo-handling machines are among the most promising equipment for electrification, being identified as central for reducing terminal emissions. Moreover, they are particularly suitable for electrification than long-distance transport applications, since charging infrastructure can be installed close to the operating area and charging can be coordinated with breaks, idle periods or shift changes. However, their adoptability largely depends on cost competitiveness such as equipment price, utilization, electricity cost, charging strategy and infrastructure deployment, while the feasibility of electrification depends strongly on the operational profile of each vehicle type, including working hours, energy consumption, battery capacity, charging power and the availability of charging windows [22].

In this context, methods to estimate the energy demand of all-electric cargo-handling equipment fleets and their smart charging operations, as developed in this thesis, are

notably important [23].

2.4 MILP-based optimization for EV fleet flexibility

Port handling's electrification increases the electricity demand of terminal operations, introducing new demand peaks. When conventionally charging, vehicles are connected as soon as they become available, or as long as their State of Charge (SOC) reaches a minimum threshold, causing uncontrolled higher peak power demand and stress on the local grid. Smart charging, by coordinating the charging process according to vehicles parameters, electricity prices and grid constraints, can address these issues [24]. Particularly, electric vehicle demand-side management can reduce peak load, improve grid stability and lower electricity costs when compared to uncontrolled charging strategies [25].

Optimization models can use different methods, among which mixed-integer linear programming (MILP) is particularly suitable when the charging problem includes both continuous variables and discrete decisions. MILP formulations can include charging power and state of charge, as well as the activation of fast charging or the selection of charging periods, together with constraints as fleet operating schedules, battery capacity or charge power limits [24]. EV charging profiles are commonly optimized using MILP, although computational performances may become an issue when dealing with large fleets [26].

The problem's objective function depends on the priorities and goals of the operator. Common objectives include peak shaving, valley filling and load shifting, minimizing the load curve or peak valley difference and improving system stability. On the other hand, cost minimization or income maximization can be also achieved [24]. In practical applications, the coexistence of different objective functions may conflict, and for this reason multi-objective or hierarchical optimization approaches, as done in this thesis, are often adopted to preserve the priority of the technical or economic purposes [27].

2.5 Distribution grid impact assessment and power-flow analysis

The electrification of port utilities as OPS and EV causes not only an increase in the general demand, but also new operational constraints for the DSO. These effects are widely discussed in the literature of electric vehicle integration, with uncoordinated charging associated with grid equipment overloading, consequently reducing hosting capacity. In this context, mitigation mechanisms are crucial and impactful. [28] [29].

Distribution grid impact evaluation is therefore fundamental to verify whether the existing network can operate within the limits. The main technical indicators usually considered are bus voltage magnitudes, line loading, transformer loading, network losses and the duration of critical events [30].

Power flow analysis is one of the standard methods used to assess these impacts, by calculating the voltage magnitudes, loading levels and active and reactive power flows through lines and transformers. In this thesis pandapower was used, already successfully applied in several grid studies [31]. The tool is suitable for this work because it allows the Oskarshamn medium-voltage network to be represented through buses, lines, transformers and time-varying loads, with the relative indicators used to compare conventional charging, optimized charging and PV/BESS integration scenarios, checking whether flexibility measures can reduce network stress, improve asset utilization and potentially defer reinforcement needs.

2.6 Research gap and contribution of this thesis

Existing literature shows how port electrification is commonly evaluated as an energy transition hub towards decarbonization and GHG emission reduction, emphasizing the role of electrification, renewable generation and storage [10]. Other works focus specifically on OPS, analysing their technical requirements and recognizing their environmental benefits [17].

Another research stream evaluates the electrification of cargo-handling equipment and general EV fleets, showing how coordinated charging can reduce peak demand, lower the costs and mitigate grid stress, as stated in Section 2.4.

However, fewer studies combine all these elements into a one, complete port-operational and grid-oriented framework. OPS demand, electric port vehicles, terminal base loads, PV generation, BESS dispatch and distribution-network constraints are often analysed separately, even though in real port operation they interact on the same electrical infrastructure, underestimating the combined impact of electrification especially when both OPS and fleet charging occur at the same period.

This thesis therefore contributes to this gap by assessing the planned electrification of the new Ferry and Cargo Terminals at the Port of Oskarshamn, combining vessel OPS demand, electrified port handling equipment, MILP-based smart charging, PV and BESS integration, and time-series power-flow analysis of the local distribution grid. In doing so, it evaluates both the operational feasibility of port electrification and its technical consequences for the DSO, including line loading, transformer loading and voltage magnitudes. Particularly, the work investigated different operating strategies, making it possible to compare distinct scenario and their impact. The thesis is consequently positioned at the intersection of sustainable port development, smart charging optimization and distribution-grid planning.

Chapter 3

Research Objectives, Questions and Limitations

3.1 Problem and Objectives

The expanding port area will require additional power from the grid to support the new infrastructure and the electrification of the totality of the utilities, raising critical concerns about the current network, which is roughly at full capacity. Does the grid need to be strengthened? How will load optimization strategies, such as EV demand-side management and peak shaving through battery storage impact the reduction of grid stress, allowing better utilization of the already installed capacity while maintaining the reliability for end-user?

This work aims to evaluate the electrical demand of the two new terminals, optimizing their flexible loads and integrating PV and battery energy storage systems BESS, hereby analyzing the robustness of the existing electrical grid through detailed simulations. The two new areas of the port will replicate the operational behavior of two already existing port zones, modelling the full electrification of current oil and gas-powered utilities and meeting the upcoming EU regulations.

The goals of this project are as follows:

- By understanding the behaviour of the two existing port areas, replicate it into the new ones, and replace the oil and gas powered utilities to electrical equivalents for an accurate baseline model.

- Integrate and model OPS operations, following FuelEU directives [15].
- Integrate RE such as PV and BESS following a techno-economic evaluation [32].
- Develop a MILP optimization tool, achieving peak shaving and mitigating grid constraints, diminishing operational and infrastructure costs for the Port and the DSO.
- Test and analyze grid robustness.

3.2 Limitations

The current work is subject to different limitations, related to the assumptions required to model the future port operations, or the availability of input data and the deterministic optimization framework. While all the assumptions have been agreed with Oskarshamn Energi and Smålandshamn AB, actual future operations may differ from what is represented in the model.

Firstly, since the analysed electrification has not yet been implemented, the vehicles operational parameters have been based on expected future conditions. In this context, charging and discharging efficiencies were considered as 100%, with no energy losses during these processes [33]. Secondly, OPS demand, which is represented through a trapezoidal load profile, has been calculated using scientific reference [34], but each period of the trapezoidal profile is assumed to be constant. This simplification does not capture short-term fluctuations and may therefore over or under estimate the ship loads. Secondly, the proposed MILP optimisation framework follows a deterministic approach, where all relevant inputs are assumed to be known in advance. In this context, where the main inputs are electricity prices, vehicle schedules, OPS energy requirements and PV generation, the idealistic planning case is believed to not be too distant from reality, where these inputs can be easily known in advance on the day before, apart from the PV generation which is subject to variable weather conditions and that would require precise forecast methods. Moreover, the optimisation model is designed to reduce the peaks and the costs related to Port operations, as also done for the BESS dispatch strategies. While this approach inherently benefits the local grid, future work could further investigate optimisation strategies explicitly aimed at minimising the impact on the distribution network, shifting the flexible loads base on the overall city demand. The approach adopted in this study was chosen to

provide comparable benefits for both the Port and DSO, with the first acting as the main investor. In this picture, ensuring clear and tangible economic benefits for its operations was considered a key requirement.

Despite these limitations, the adopted modelling approach provides a consistent base to assess the electrification of the port, and investigate the interactions between port operations, charging flexibility, local energy resources and distribution network.

Chapter 4

Methodology

This study applies a quantitative, simulation based methodology to assess the technical feasibility and grid performance associated to the construction of two new terminals at the Port of Oskarshamn. The objective it is to evaluate whether the existing electrical grid is robust enough to accommodate the increase in demand, identifying possible operational constraints and applying MILP techniques to optimize load behaviour and integrate PV and BESS.

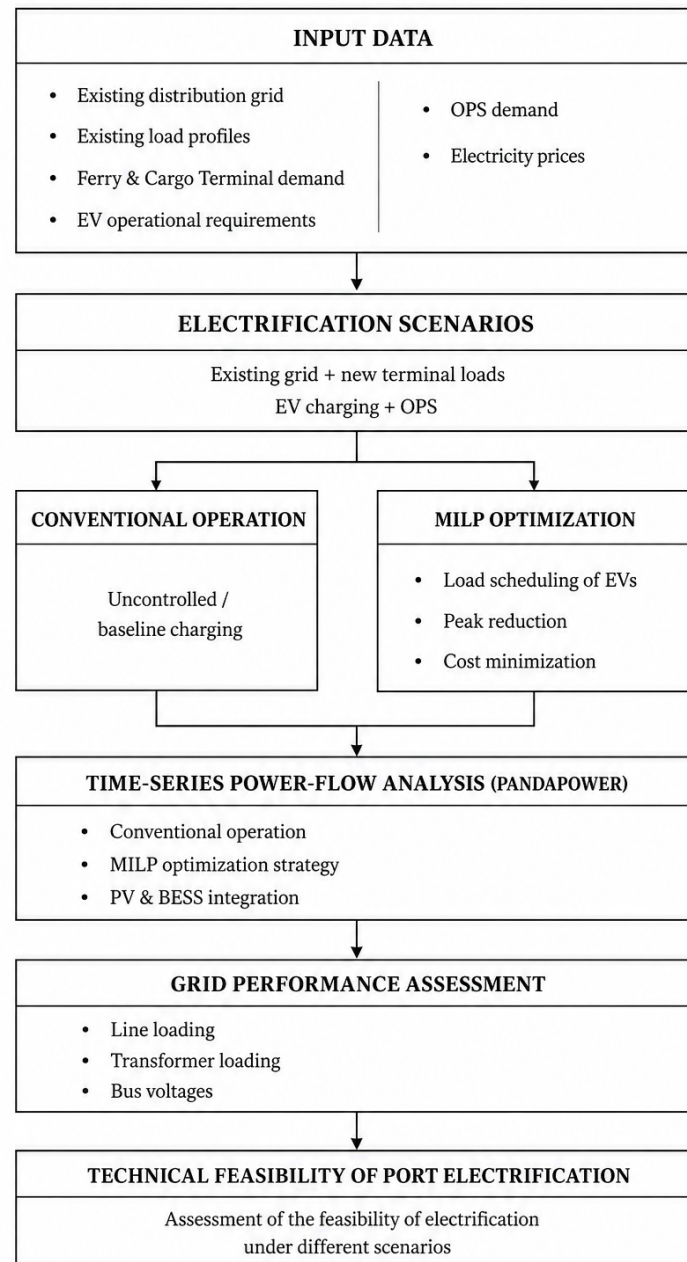


Figure 4: Overview of the methodological framework adopted for the assessment and optimization of the Port of Oskarshamn electrification.

4.1 Data collection and preprocessing

The data collected are related to the current grid and port's operations and existing infrastructure.

4.1.1 Load Data

Current base-demand (Ampere) of the existing ferry and cargo terminals was obtained from Oskarshamn Energi, specifically representing the period from March 2025 to February 2026 with a 15 minutes resolution. The base load stands for the aggregated consumptions of the outdoor lighting, telecom and data, cranes demand and other general electrical systems. This data was then converted to kW and kVAr multiplying the Ampere values to 10 kV, 0.95 (power factor) and $\sqrt{3}$ (3-phase system). Only ferry terminal base power demand was scaled up of a 5000/1630 factor, representing the difference between the existing and new area in m^2 , considering the increase of area directly proportional to the increase of lighting and other general electrical systems demand. Finally, for simplicity and with no major changes happened during the last year, January and February 2026 data were shifted at the beginning of 2025, in order to complete a full 2025 year with a total of 35,040 fifteen minutes time-steps.

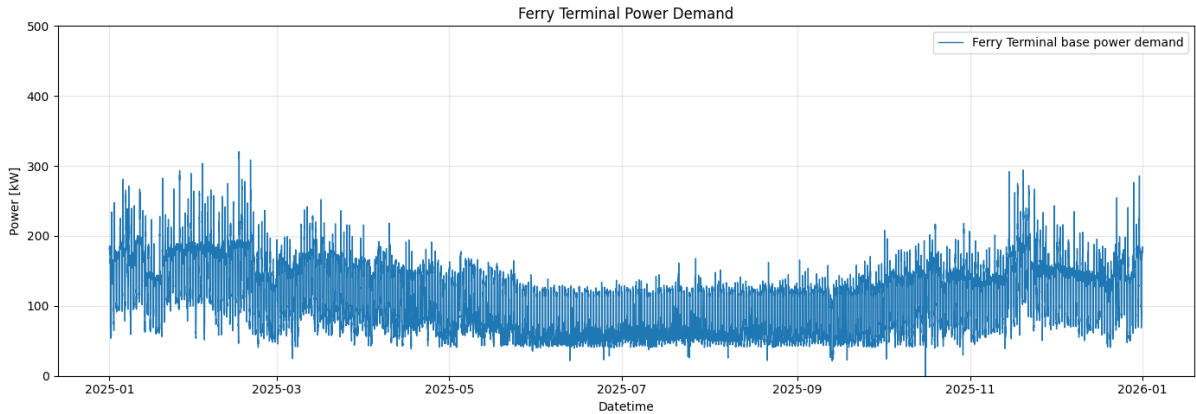


Figure 5: Base load demand

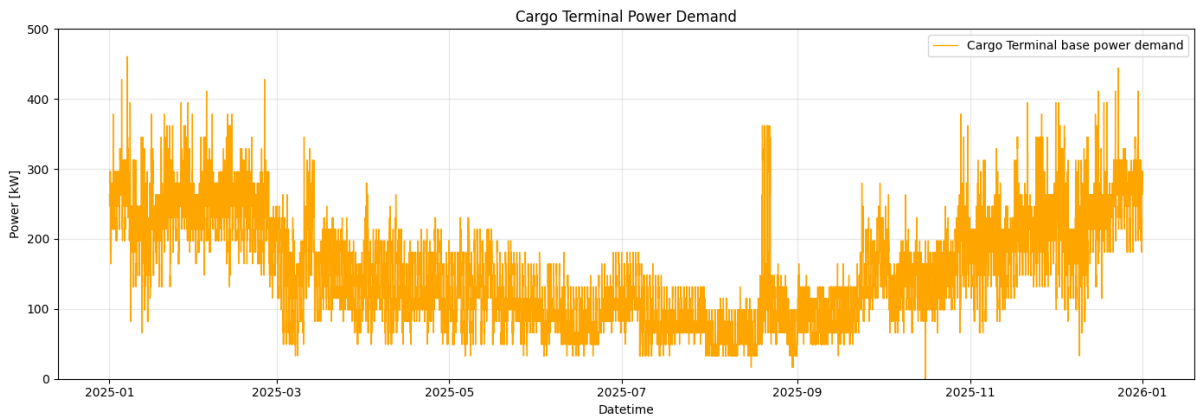


Figure 6: Base load demand

4.1.2 Electrification and future load assumptions

EV characteristics

Part of the aim of work was to evaluate the future electrical demand of the Port after phasing-out all fossil fuels powered utilities. Hereby, all the ICE vehicles were replaced by equivalent EV. The EV's technical parameters were collected from [35], and are shown in the table below.

Table 1: Specifications of EV models

Model	Category	Battery Capacity (kWh)	Discharge rate (kWh/h)	Fast Charge Capacity (kW)
Merlinum EA 6-600	Forklift	64	14	128
SMV 16 Ever	Forklift	110	20	147
Volvo L120H	Loader	237	47.5	158
SMV 46/32 E-ver	Reach Stacker	276	50	276
ERG450-65CS	Reach Stacker	587	115	587
Terberg RT 223 EL.	Terminal Mover	150	22.5	200

Of the vehicles listed in Table 1, 3 Terberg RT 223 EL. terminal movers, 3 Volvo L120H loaders, 2 Merlinum EA 6-600 forklifts and 4 SMV 16 E-ver forklifts are assigned to the Ferry Terminal, while 2 Merlinum EA 6-600 forklifts, 6 SMV 16 Ever forklifts, 6 Volvo L120H loaders, 2 SMV 46/32 E-ver reach stackers, 1 ERG450-65CS reach stacker and 3 Terberg RT 223 EL. terminal movers operate in the Cargo Terminal area, in accordance with Smålandshamn AB instructions.

OPS modelling

OPS activities were modelled using the installed auxiliary engine power (kW) of the vessels arriving at the port, together with their relative auxiliary engine load factor

[34].

Table 2 summarizes all vessel categories that berthed at the two terminals during 2025. Although bulk carriers and the general cargos weight less than 5,000 GT, and would therefore be exempt from OPS requirements (FuelEU), they are nevertheless included in the analysis. This choice is motivated by the uncertainty associated with the assumed vessel sizes and by the adoption of a conservative approach.

Table 2: Specifications of OPS vessel demand

Vessel	GT (ton)	Installed aux engines (kW)	Load factor	OPS power demand (kW)
Oil/Chemical Tanker	7479	900	0.67	603
Bulk carrier	4216	596	0.22	131
General cargo	4497	522	0.22	115
Visby	32447	4974	0.30	1492
Gotland	32477	4979	0.30	1494
Drotten	29764	4560	0.30	1368

Together with the arrival and departure schedule of each type of ship, provided by the Swedish Maritime Administration (SMA), it was possible to dimension the OPS demand of each ship, and so the power supplied by the port. To enhance realism and avoid extremely rapid peaks, power is scaled with ramp multipliers: short stays ($x \leq 4$ hours) use 85% of rated demand at the first and last step, while long stays use 42.5% and 85% at the first and last and at the second and second-last steps, respectively. For the Ferry Terminal, the SMA dataset indicates the simultaneous presence of two vessels on only a single occasion over the all year, likely being an outlier or the result of some maintenance. This event was treated as follows: when more than one ship is berthed at the same time at the Ferry Terminal, only the most recently arrived counts, as if only one berth connection is active.

The OPS dimensioning results for the two terminals are shown in Figure 7 and 8. Figures can be better evaluated by looking at Table 3, where the load factor is the ratio of average OPS demand to peak OPS demand and the utilization factor is the share of timesteps with demand greater than zero.

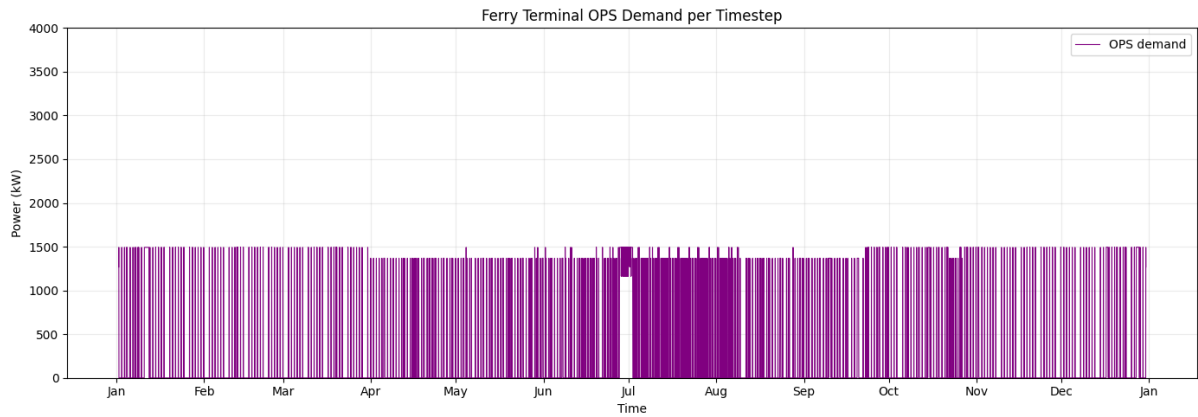


Figure 7: Ferry Terminal OPS demand

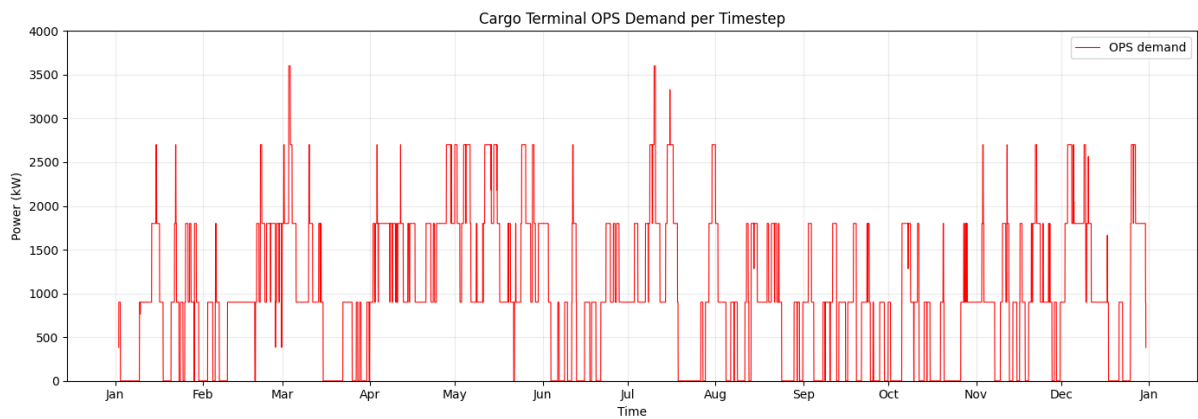


Figure 8: Cargo Terminal OPS demand

Table 3: Demand characteristics of the evaluated terminals

Metric	Ferry Terminal	Cargo Terminal
Average demand	150 kW	992 kW
OPS peak demand	1.49 MW	3.60 MW
OPS load factor	10.1%	27.6%
OPS utilization factor	11.0%	72.4%

4.1.3 Electrical network

The electrical infrastructure of the city of Oskarshamn consists of a medium voltage distribution network operating at 10 kV. The city is supplied by E.ON, providing electricity at 50 kV to two substations. Particularly, Alvarsberg (ALV) substation is linked to the existing terminals and converts the electricity supplied by E.ON to 10

kV. Along the network, several transformers are placed to fulfill the LV electricity requirements of Oskarshamn Energi clients, stepping down the voltage to 0.4 kV.

Network data were collected both from Oskarshamn Energi and [35]. Changes were related to the new infrastructure needed to connect the two new terminals to the grid.

The electrical infrastructure of the city of Oskarshamn was modeled as a medium voltage distribution network using a bus-branch representation, where buses represents electrical nodes and branches are lines and transformers. Designing the internal grid of the Port was not in the scope of the work.

Particularly relevant buses for the analysis are Bus_0 , representing ALV, and LIH_{mv} and KLU_{new} , representing the connection point between the grid and the new Ferry and Cargo terminals, respectively. Transformers LIH and KLU_{new} convert the 10 kV electricity to 0.4 kV: OPS were connected to the 10 kV side of the buses, while base load demand and EV charging operations were connected to the 0.4 kV side.

Lines were parametrized according to their (km) and resistance (Ω/km), reactance (Ω/km), capacitance (nF/km) and maximum current (kA) values, while transformers were modelled based on their rated apparent power, nominal voltage levels, and short-circuit impedance (Appendix B); loads were represented as constant power (PQ) elements with fifteen minutes time-varying active and reactive power demand.

4.1.4 Day-ahead electricity prices

Day-ahead electricity prices were downloaded by ENTSO-E and for Sweden SE3 bidding zone for the whole 2025 year [36]. Since the EU's day-ahead 15-minute Market Time Unit (MTU) began on September 30 2025, for the period prior to this date, a realistic quarter-hour variability was introduced, generating small correlated random deviations with a magnitude of about 2% of the hourly level, finally re-centring the four deviations to zero mean, unchanging the hourly average and preserving the original price while creating a realistic 15-minutes difference.

Electricity cost for the Port has been calculated considering that the spot price has a loss coefficient of 5.61%; aggregated to the spot price is a fixed price of 2.505 öre/kWh on the total consumption per month, as provided by Oskarshamn Energi.

4.2 Charging strategies

Two different charging strategies were implemented and distinctively applied to the Ferry and Cargo terminals: one to simulate a conventional-base charging pattern pursued by the port’s employers and EV utilizer, which lately serves as a baseline comparison scenario; the second one is the optimized charging management strategy.

Consumptions profiles of each vehicle were daily synthetically generated, considering allowed working windows (from 7 AM to 7:30 PM, breaks showed in Table 4 excluded) and total working hours per vehicle per day (Appendix A.1). SOC is lower bounded to 10% to prevent complete battery discharge and preserve battery health, and upper bounded to 100% to realistically avoid potential disruptions to Port operations in the event of unforeseen last-minute requirements.

Common inputs and constraints between the two models are listed in Table 4, and were agreed upon with Oskarshamn Energi and Smålandshamn AB.

Table 4: Summary of rule-based charging model parameters

Parameter	Description
Time horizon T	Full-year timeline for 2025 at 15-minute resolution (35,040 timesteps).
Vehicle set V	Ferry and Cargo Terminal’s fleet vehicles.
Consumption profile	Daily vehicle energy consumption aligned to the common 15-minute simulation timeline
SOC limits	SOC of each vehicle is bounded between 10% and 100%.
Charging windows	Charging is available during non-working time periods, particularly from 9 to 10 AM, from 12 to 1 PM, from 3 to 4 PM and during night time from 8 PM to 7 AM. Only ERG450-65CS Reach Stacker is enabled to charge whenever is not operating, due to its high discharge rate.
07:00 checkpoints	Daily 07:00 timestamps enforced across the entire simulation year.
Slow charging constant P_{slow}	Fixed slow-charging power equal to 22 kW.
Fast concurrency limit	Maximum number of simultaneously fast-charging vehicles equal to 4 in the Ferry Terminal and 5 in the Cargo Terminal, for a total of 9 stations.

The two strategies are described below.

4.2.1 Conventional charging

The charging problem is formulated on a 15 minutes resolution, receiving as inputs the vehicle specific battery capacities, fast and slow charging limits and historical vehicle energy consumptions. In this case, the charging dispatch is implemented through a deterministic rule-based scheduling approach.

Additional heuristic rules are included to assure operational realism. For the Ferry Terminal, an intraday skip-charging rule prevents vehicles having more than 50% SOC and that can complete their daily operation to charge. In parallel, the forward looking "fully charged by 7:00 AM" rule prevents possible undercharging patterns. Related to the Cargo Terminal, the "over 50% skip fast-charge" rule is not used, due to the stricter boundaries given by the type of vehicles, while only ERG450-65CS (Truck 444 Krokmaskin) follows independent rules, dictated by its high battery capacity and consumptions.

Overall, this model represents a realistic benchmark to test the improvements brought by the optimization framework.

4.2.2 Optimization Model and Solver

The optimization framework developed in the notebook models the charging strategies of the ferry and cargo terminal EV fleets, utilizing a MILP algorithm, implemented in Pyomo and solved with Gurobi at 15 minutes resolution over the 2025 entire year. The framework integrates multiple data sources in a unified time structure, as base terminals electricity demand, shore power demand from berthed vessels, vehicle specific battery capacities, fast and slow charging limits as well as historical vehicle energy consumption and 15 minutes day-ahead electricity prices for area SE3 in Sweden.

The optimizer determines the charging power profile of each vehicle, the evolution of its SOC and the activation of fast charging. The model is constrained among battery capacity limits, inter-temporal SOC dynamics, charging availability windows depending on the working schedule, fast charging available spots (4 in the Ferry Terminal, 5 in the Cargo Terminal), and a hard operational requirement that all vehicles must reach full charge at 07:00 checkpoints to ensure full working reliability.

The solution is obtained through a four-stage lexicographic optimization procedure, preserving the hierarchy between objectives, identified in minimizing peak power demand and total charged energy, while penalizing abrupt charging ramps, deviations from a reference charging profile, charging during ship-presence intervals, and excessive high-power charging, finally choosing the best solution that also minimizes electricity procurement costs.

The described methodology results in a realistic and implementable charging schedule that balances peak shaving, load shaping, and DSO and Port's cost efficiency under terminal and fleet operating conditions.

Table 5: Summary of the MILP-based charging optimization framework

Aspect	Description
Optimization type	Mixed-Integer Linear Programming (MILP) implemented in Pyomo and solved using Gurobi.
Time resolution	15-minute time step over a full-year simulation horizon.
Integrated inputs	Fixed terminal demand, OPS demand from passenger vessels, EV battery capacities, charging limits, historical vehicle energy consumption, and day-ahead electricity prices.
Decision variables	Vehicle charging power trajectories, state-of-charge evolution, and fast-charging activation.
Operational constraints	Battery capacity limits, inter-temporal SOC dynamics, charging availability windows, prohibition of charging during operation, simultaneous fast-charging limits, and mandatory full charge at 07:00 checkpoints.
Peak demand modeling	Total peak demand modeled as the maximum combined terminal load and fleet charging demand.
Stage 1 objective	Minimization of total peak power demand.
Stage 2 objective	Minimization of total charged energy while preserving the optimal peak demand.
Stage 3 objective	Minimization of charging-profile smoothness penalties, including charging ramps, deviations from a reference profile, charging during ship presence, and excessive high-power charging.
Stage 4 objective	Minimization of electricity procurement costs while maintaining previously optimized objectives within predefined tolerances.
Optimization strategy	Four-stage lexicographic optimization preserving the hierarchy between technical and economic objectives.
Final outcome	Generation of realistic and implementable charging schedules balancing peak shaving, load shaping, operational feasibility, and cost efficiency.

4.3 Grid impact of PV and BESS

A relevant part of the work was evaluating the impact of the integration of optimally sized PV and BESS in both the Ferry and Cargo areas. The system design was carried out by [32], in coordination with the present Master thesis work.

The optimal system design, shown in Table 6, differentiate the cost minimization and peak minimization scenarios, where:

- **MILP Cost Minimization** objective function:
MAX Net Present Value (NPV) + MAX Self-Sufficiency Ratio (SSR)
- **MILP Peak Reduction** objective function:
MIN Capital Expenditure (CAPEX) + MAX Peak Reduction

Moreover, [32] analysis provided a 15 minutes resolution battery energy dispatch profile, utilizing the same inputs and EV optimized charging profile, and aligned with the temporal resolution adopted in this work, which was subsequently utilized to evaluate the impact of the system on the grid.

Table 6: Optimal PV and BESS configurations for terminal electrification [32]

System Configuration	PV (kWp / GWh)	BESS (kWh / kW)
Passenger Terminal		
PV + BESS (Min. Cost)	900.9 / 0.8	849 / 310
PV + BESS (Max. Peak Red.)		1741 / 435
Cargo Terminal		
PV + BESS (Min. Cost)	2503.8 / 2.2	1904 / 803
PV + BESS (Max. Peak Red.)		4969 / 1242

The grid impact evaluation related to the different scenarios and presented in Chapter 5, is referred to the PV + BESS peak reduction configuration. Results and consideration on the cost minimization configuration can be found in Appendix C.3.

4.4 Power flow analysis

Power flow analysis is a technique used to evaluate, by simulating electrical networks operations under steady-state conditions, key electrical variables such as bus voltages, line currents, transformer loadings and system losses. It helps in understanding the

electrical power flow from the generation sources to the connected loads, thereby providing useful information of overall system performance. Moreover, by identifying voltage or current violations, power flow support the safe and reliable operations of electrical networks.

For this study, pandapower tool has been used. Pandapower is an open source Python based tool providing an high-level interface for creating network models, performing power flow simulations and helping to identify any potential issues or areas for improvement.

4.5 Results evaluation

Simulation results of the Ferry and Cargo terminals were evaluated considering both Oskarshamn Energi and Smålandshamn AB perspectives, through the following Key Performance Indicators (KPI):

- Load power demand peaks (kW)
- Total EV's charged energy (kWh)
- Total EV's charging costs (SEK)
- Line loading percentage (%)
- Transformer loading percentage (%)
- Bus voltage magnitude (p.u.)

While the Ferry Terminal's grid impact analysis is highly relevant for its future construction, the analysis of the Cargo Terminal primarily serves to assess the (un)feasibility of connecting it to the evaluated side of the network. This is because the new Cargo Terminal will be connected to an other side of the grid, yet to be constructed and consequently analyzed. Nonetheless, the EV charging optimization strategies are in both case highly relevant and can be independently applied.

Among the several simulations and analysis pursued, it was selected the approach giving the best results in terms of grid impact mitigation and consequently reduction of infrastructural costs for the DSO, followed by the minimization for Port's operational costs.

Chapter 5

Results

This section shows the results obtained from the various simulations and analysis on both the Ferry and Cargo terminals. For each terminal, it will be presented the EV charging results, their impact on the grid, and how the integration of a BESS and PV system influenced the load.

5.1 Ferry Terminal

5.1.1 Conventional operation scenario

The conventional system comprehends the base load, the OPS demand, and the EV charging demand dimensioned as explained in Section 4.2.1. The EV demand is in this case particularly relevant, as it will serve as a baseline to then evaluate the optimization strategy.

Figure 9 shows the EV demand during the year.

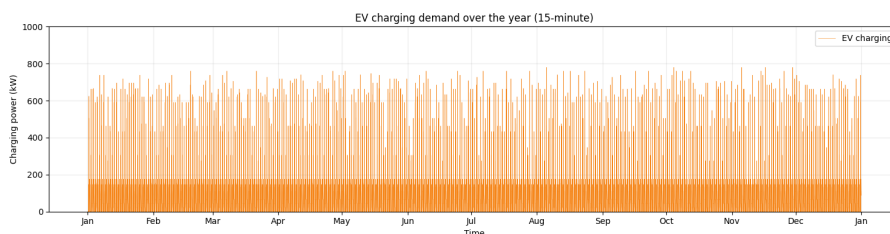


Figure 9: Conventional Ferry Terminal EV demand

Due to the highly dynamic EV schedule (and consumptions) pattern, as well as the OPS

demand and terminal base load, the resulting charging demand exhibits significant day-to-day fluctuations. To facilitate a clearer evaluation, Figure 10 shows the week from October 13 to October 19, selected as a representative case study. This period corresponds to the largest electricity price range during the year, ranging from 0.09 to 5.58 SEK/kWh, and is therefore particularly suitable for illustrating the interaction between charging demand and market conditions.

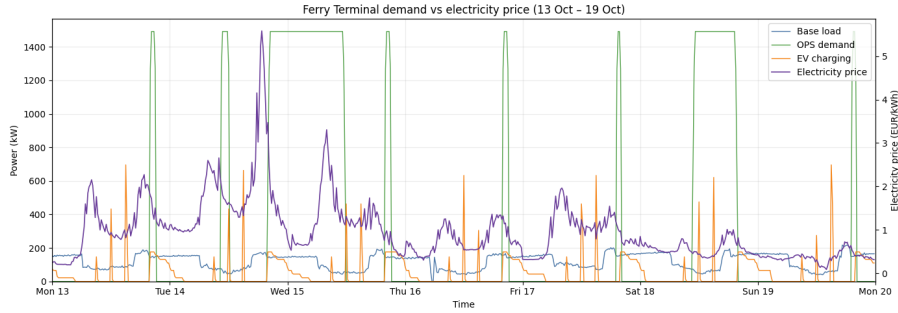


Figure 10: Ferry Terminal demand and electricity prices

In this case, the charging pattern does not consider price fluctuations or peak-demand minimization. As shown by the gradual increase in charging power during the intraday charging windows, and to preserve realism, vehicles are charged only when their state of charge (SOC) falls below 50%.

Finally, to provide a comprehensive reference, Figure 11 and 12 present the annual load duration curves, obtained by sorting the total demand values of each time step in descending order.

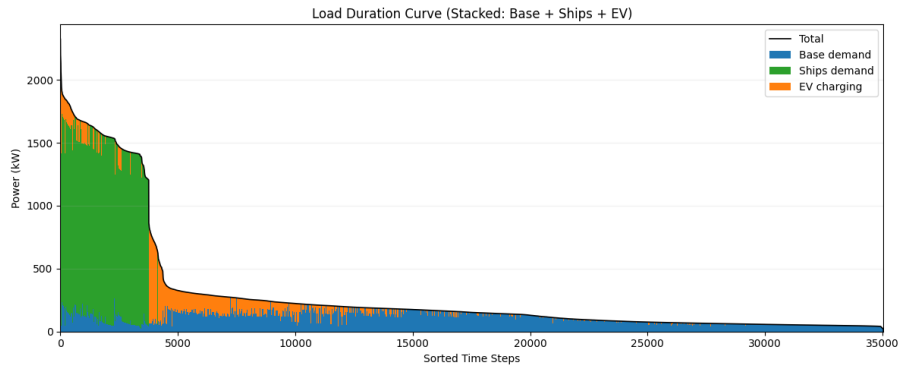


Figure 11: Ferry Terminal sorted energy demand (conventional EV charging)

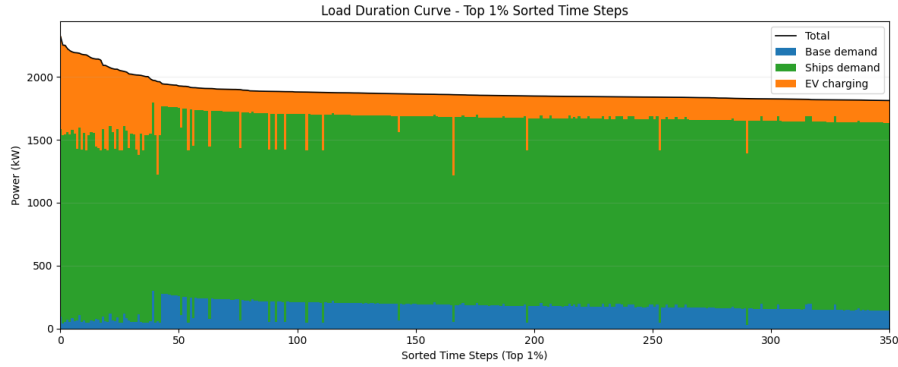


Figure 12: Ferry Terminal top 1% sorted energy demand (conventional EV charging)

All the most relevant KPI are summarized in Table 7.

Table 7: Conventional EV charging results

Metric	Value
Peak demand without charging	1.80 MW
Peak demand with charging	2.32 MW
Peak demand increase	526.3 kW
EV charging demand at system peak	738.0 kW
Total charged energy	398,065 kWh
Total charging cost	213,173 SEK

Grid Impact

Grid impact evaluation of the conventional charging scenario demonstrates how adding uncontrolled loads as the EV charging, and highly demanding OPS operations, significantly impacts the network. This happens both on the network's closest part to the new port loads (bus *LIH*), as well as on the overall grid.

When evaluating the line loading percentages, results (Figure 13) show that lines close to the load remain below the 80% load threshold, while the cables at the beginning of the grid and responsible of power transmission from ALV (main substation) to the buses, suffer of sever overloading (Figure 14)

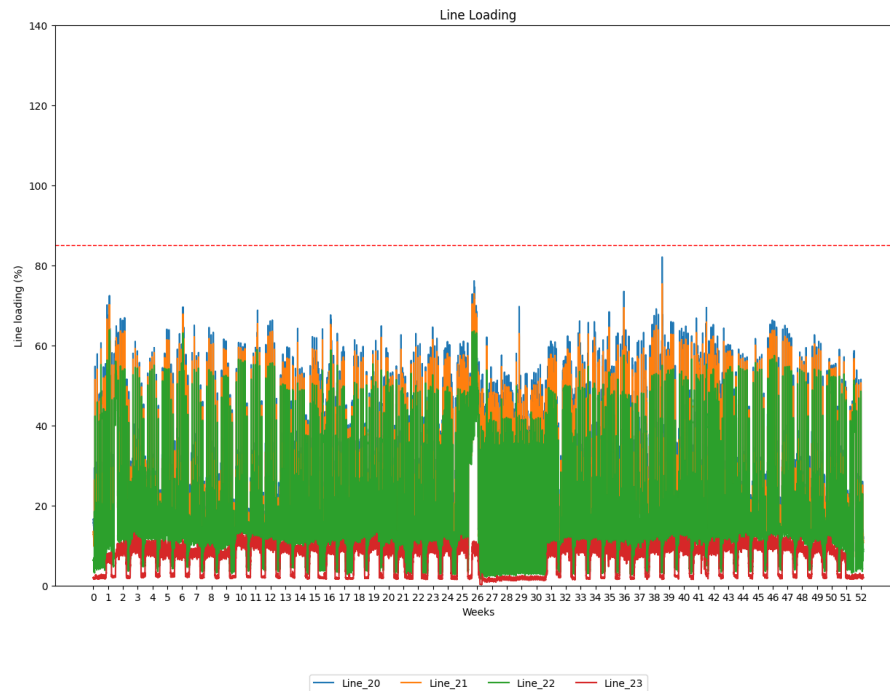


Figure 13: Loading percentage of the lines closest to the load (conventional EV charging)

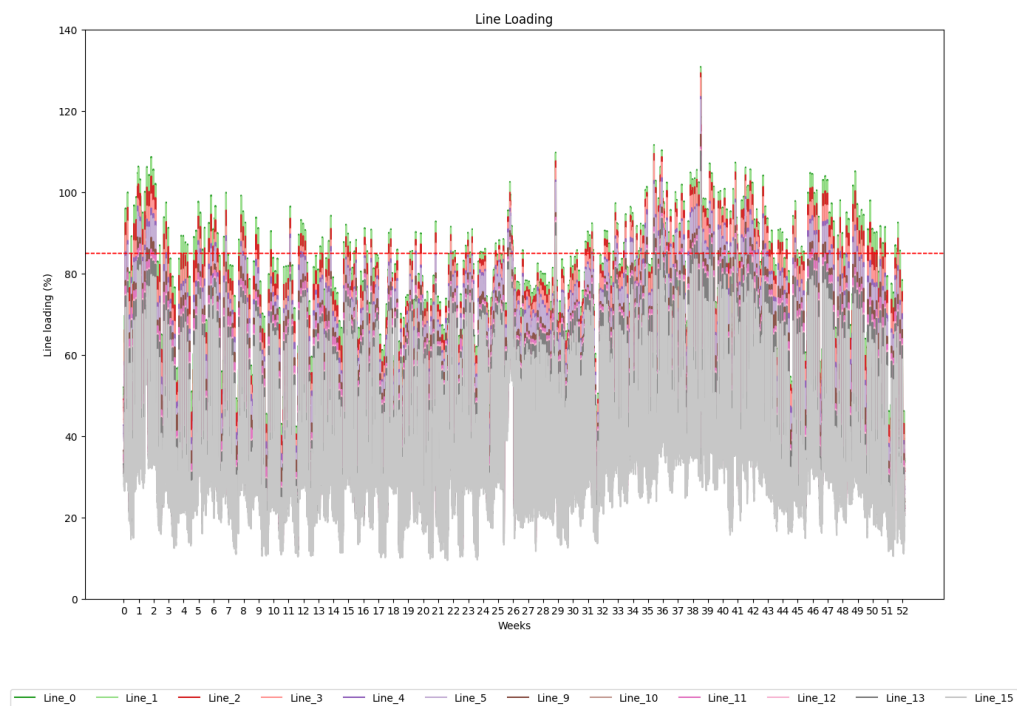


Figure 14: Loading percentage of main transmission corridor lines (conventional EV charging)

All the lines represented in Figure 14 are subject to overloading higher than 100%, with lines 10, 11, 12, 13, 15 that exceed the limit value only for 15 minutes during the whole

year. The optimization scenario and the BESS and PV integration goal comprehend the reduction of the max loading percentage, as well as the total 15 minutes time-steps over the thresholds, consequently lowering consecutive time-steps above the limits. Particularly, thermal limit for lines (computed as line loading by pandapower) is not an absolute limit, and for short periods of time can be retained acceptable [37].

Table 8 collects the overloading characteristics of the most influential lines of the network.

Table 8: Line loading statistics under the conventional operating scenario

Line	Max Loading (%)	15m steps $\geq 80\%$	15m steps $> 100\%$	Max Consecutive steps $> 100\%$ (h)
Line_0	130.88	1796	158	19 (4.75)
Line_1	130.54	1750	146	19 (4.75)
Line_2	129.34	1450	82	19 (4.75)
Line_3	128.14	1176	40	19 (4.75)
Line_4	123.51	874	25	19 (4.75)
Line_5	122.74	791	25	19 (4.75)
Line_9	114.20	389	3	2 (0.50)
Line_10	111.10	265	1	1 (0.25)
Line_11	111.10	265	1	1 (0.25)
Line_12	110.55	186	1	1 (0.25)
Line_13	110.13	163	1	1 (0.25)
Line_15	105.06	57	1	1 (0.25)
Line_20	82.09	1	0	0 (0.00)
Line_21	75.41	0	0	0 (0.00)
Line_22	63.94	0	0	0 (0.00)

The study also considers the impact on the LIH transformer, serving the base load and the EV charging, and the bus voltage magnitudes, while OPS demand is fulfilled at medium-voltage (LIH_{mv}) [18]. Their yearly distribution is shown in Figure 14 and 15, respectively.

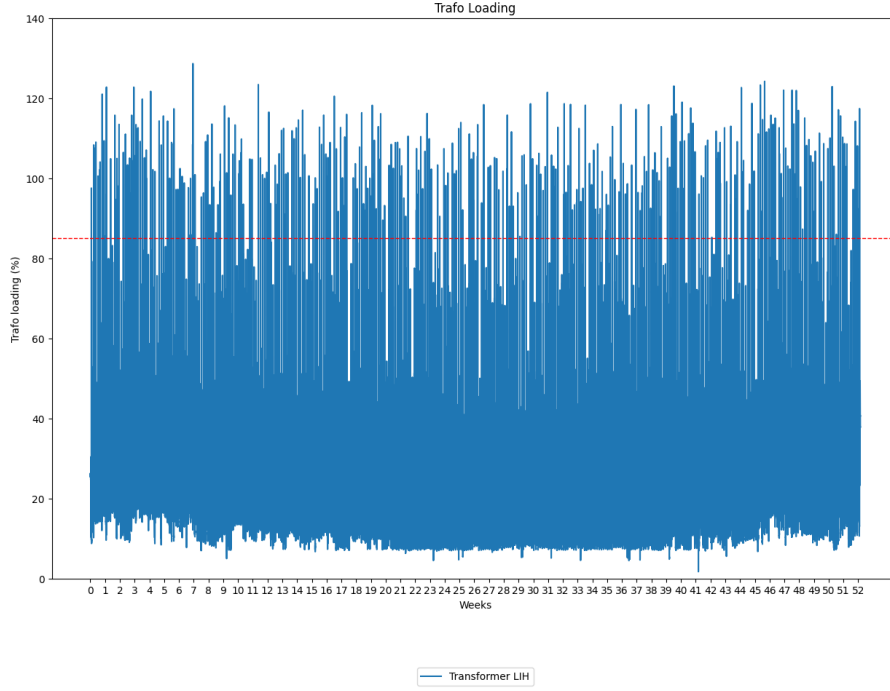


Figure 15: LIH transformer loading percentage (conventional EV charging)

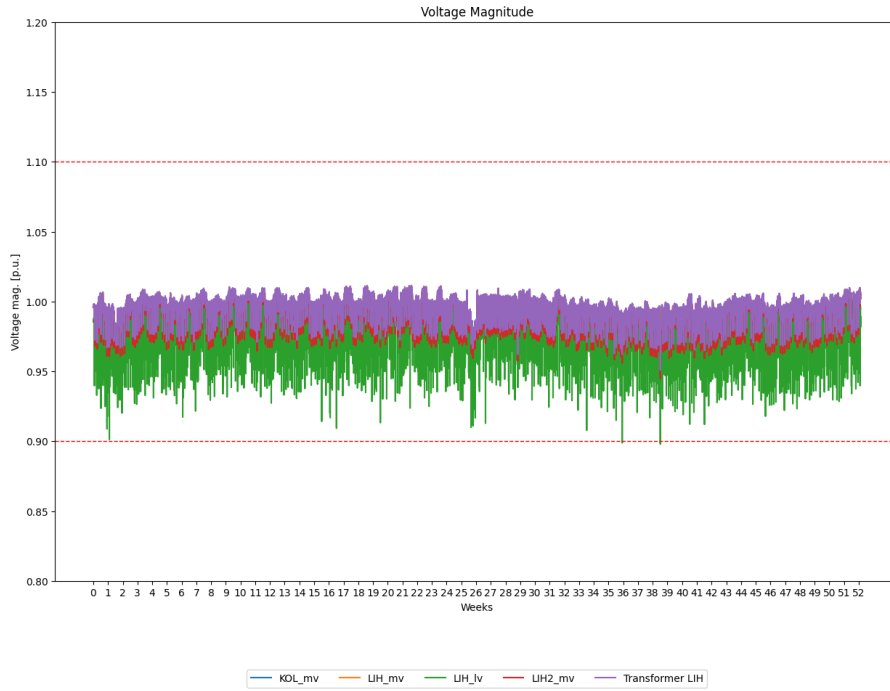


Figure 16: Voltage magnitudes of buses closest to the load (conventional EV charging)

As observed in Figure 15 and 16, the (LIH_{lv}) bus, which is the connection point for both the terminal base load and the EV charging demand, experiences the most critical operating conditions in the network, reaching bottom values close to 0.9 p.u., whereas the remaining grid buses remain comfortably within the acceptable operating limits.

On the other hand, *LIH* transformer reaches peaks of 128.67%, exceeding 80% and 100% of its rated capacity for 403 and 254 time steps during the year.

5.1.2 Optimized charging scenario

The optimized charging scenario implements the framework explained in Section 4.2.2.

Firstly, the EV optimized demand pattern is shown in Figure 17, compared to what was represented in Figure 9.

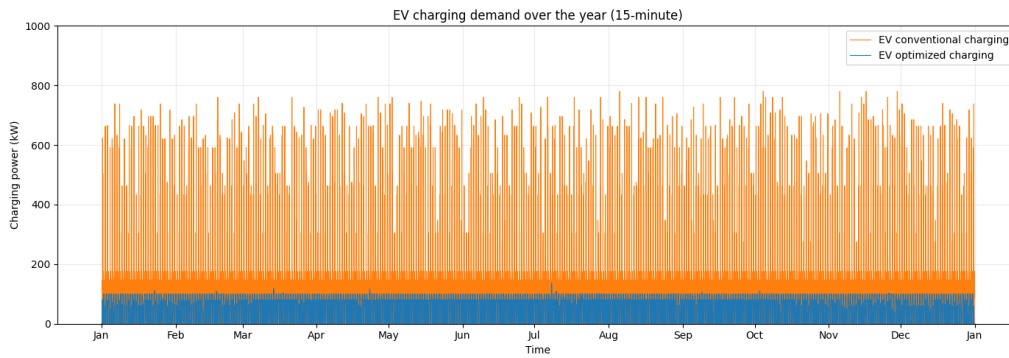


Figure 17: Optimized vs conventional Ferry Terminal EV demand)

As can be noticed, the peaks are significantly reduced (roughly 80%), and the general pattern is flattened, thus avoiding fast ramp-ups and downs causing stress to the grid.

This effect is further illustrated during the representative week October 13 - October 19 (Figure 17), where the charging profile becomes noticeably smoother, with reduced peaks and more balanced energy allocation between intraday and overnight periods, as well as shifting the charging activity away from periods with higher electricity prices. Generally, as illustrated in Appendix ??, the optimizer shifts the charging to more price-convenient time-steps only as a last priority, in order to avoid critical ramp-ups and downs. This influences what is seen in Figure 18, where it is noticeable how charging is avoided during more expensive hours, but is not highly prioritized during low prices time-steps.

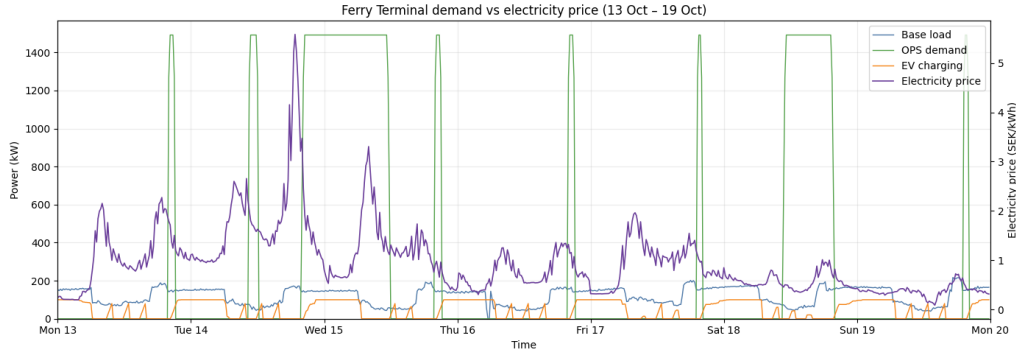


Figure 18: Optimized Ferry Terminal demand and electricity prices

Finally, as observed in Figure 19 and 20 and summarized in Table 9, the optimization strategy had a substantial impact on reducing flexible peak demand throughout the year. While OPS remained the dominant contributor to the overall load, the proposed framework shifted EV demand to more favourable time-steps, characterized by lower electricity prices and reduced network loading, thus leading to lower peaks and charging costs while preserving operational constraints.

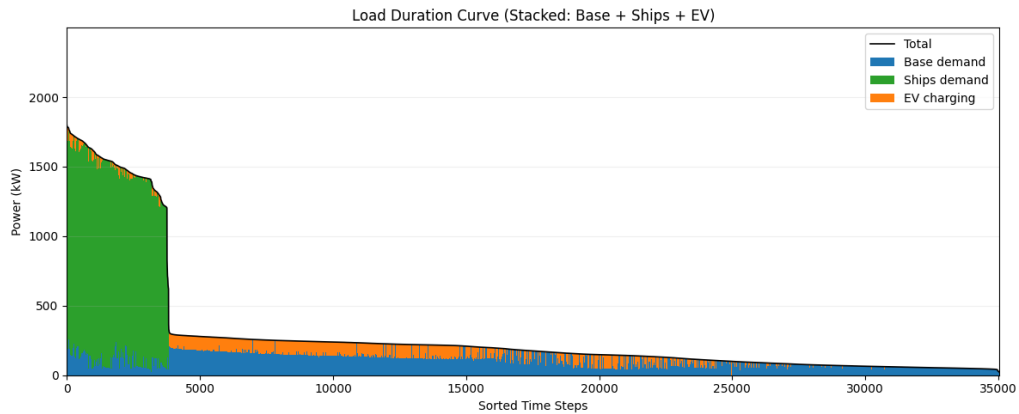


Figure 19: Ferry Terminal sorted energy demand (optimized EV charging)

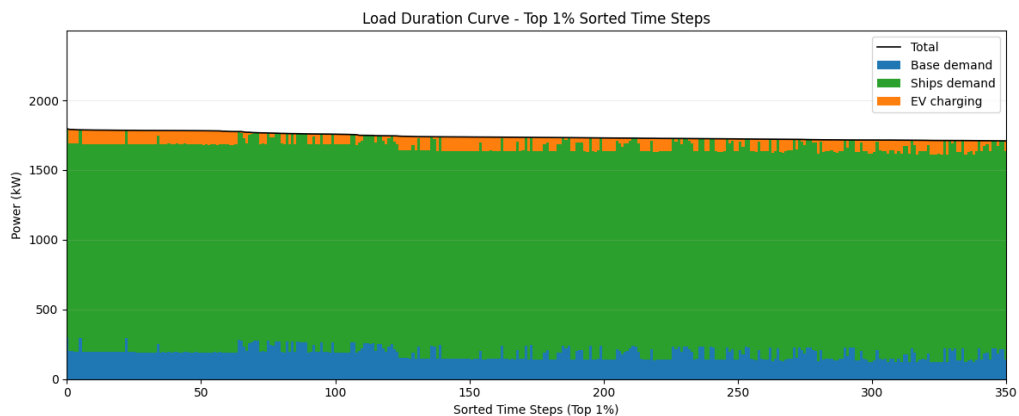


Figure 20: Ferry Terminal top 1% sorted energy demand (optimized EV charging)

Table 9 compares conventional and optimized strategies KPI.

Table 9: Comparison between conventional and optimized EV charging scenarios

Metric	Conventional	Optimized	Variation (%)
Peak demand without charging	1.80 MW		N.A.
Peak demand with charging	2.32 MW	1.80 MW	-22.7%
Peak demand increase	526.3 kW	0.0 kW	-100.0%
EV charging demand at system peak	738.0 kW	0.0 kW	-100.0%
Total charged energy	398,065 kWh	397,220 kWh	-0.2%
Total charging cost	213,173 SEK	174,482 SEK	-18.1%

Grid Impact

The optimized framework aims to solve the problems noticed in the conventional operation scenario.

Firstly analysing the lines, it can be observed a general decrease in the cables loading percentages, with Line 0, which is the first one of the network and so responsible of the all power flow, reaching a maximum value of roughly 118.60%, compared to the 130.88% of the conventional scenario. On the other hand, the lines closer to the Port's connection point (lines 20, 21, 22), also faced a significant peak load reduction of roughly 10-15%.

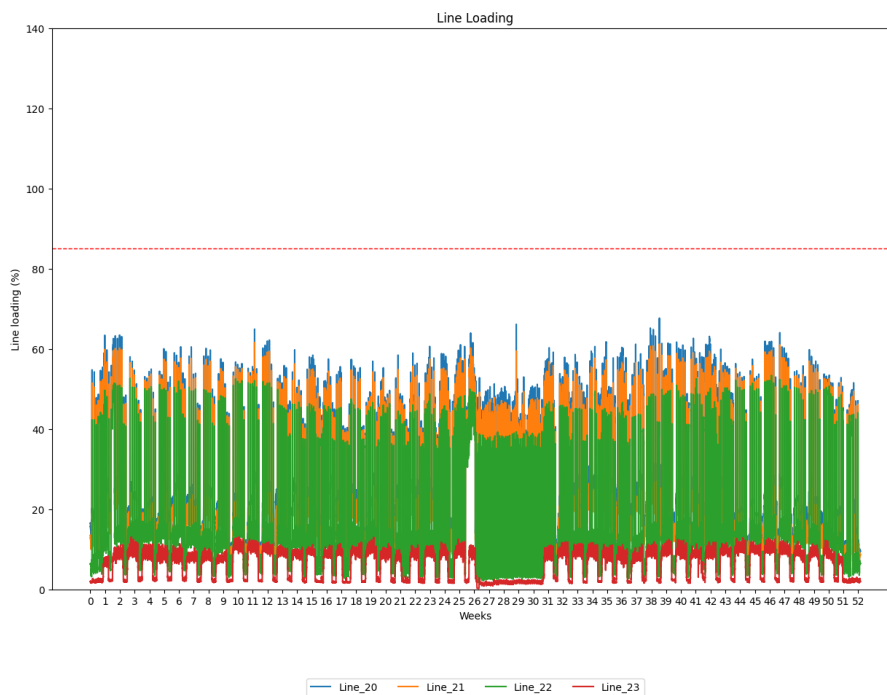


Figure 21: Loading percentage of the lines closest to the load (optimized EV charging)

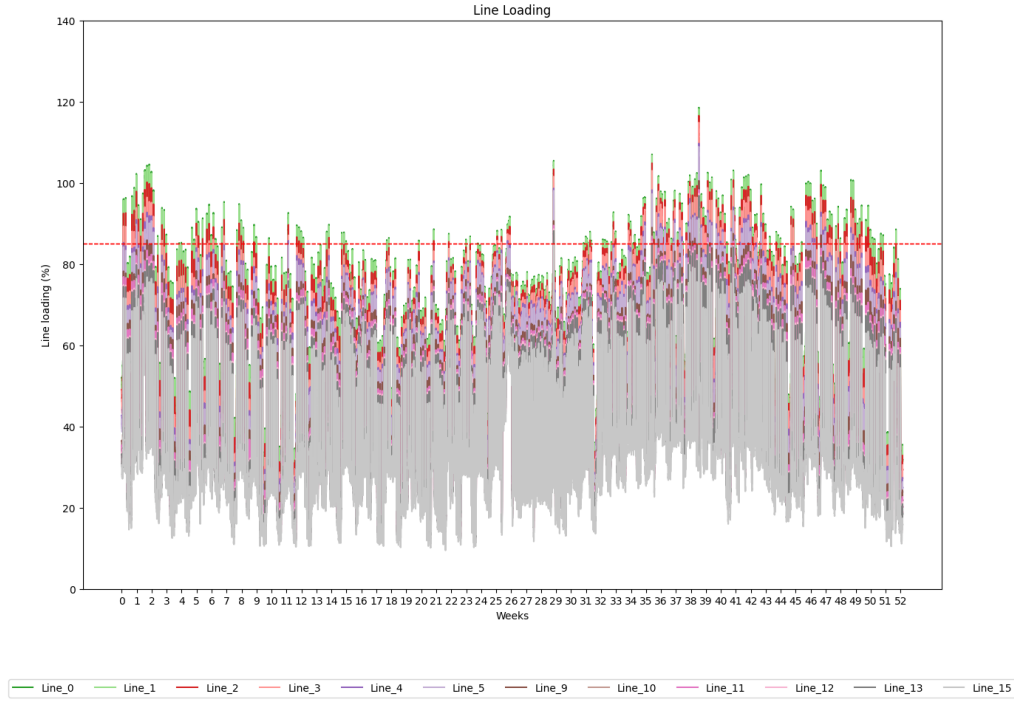


Figure 22: Loading percentage of main transmission corridor lines (optimized EV charging)

Table 10 summarizes the line loading results, as well as comparing them with the conventional charging scenario, where values in brackets indicate the percentage variation with respect to the conventional charging pattern. As can be seen, the reduction is significant and spread all over the network, especially when looking at the maximum loading percentage and the reduction in terms of time-steps over the limits. However, it was not possible to reduce the maximum consecutive time-steps over 100% loading, which occur during the 29th of September, from 15:00 to 19:30. During this period, EV charging demand is limited to 15:00 to 16:00, and never goes above 40 kW, while it is present a high contribution of OPS demand, reaching peaks of 1.494 MW and significantly contributing to lines overloading.

Table 10: Line loading statistics under the optimized charging scenario

Line	Max Loading (%)	15 min Steps $\geq 80\%$	15 min Steps $> 100\%$	Max Consecutive Steps $> 100\%$ (h)
Line_0	118.60 (-9.4%)	1601 (-10.9%)	90 (-43.0%)	19 (4.75)
Line_1	118.27 (-9.4%)	1551 (-11.4%)	83 (-43.2%)	19 (4.75)
Line_2	116.59 (-9.8%)	1229 (-15.2%)	33 (-59.8%)	19 (4.75)
Line_3	114.91 (-10.3%)	964 (-18.0%)	24 (-40.0%)	19 (4.75)
Line_4	109.74 (-11.1%)	702 (-19.7%)	19 (-24.0%)	19 (4.75)
Line_5	108.97 (-11.2%)	594 (-24.9%)	19 (-24.0%)	19 (4.75)
Line_9	100.57 (-11.9%)	253 (-35.0%)	1 (-66.7%)	1 (0.25)
Line_10	98.32 (-11.5%)	144 (-45.7%)	0 (-100.0%)	0 (0.00)
Line_11	98.32 (-11.5%)	144 (-45.7%)	0 (-100.0%)	0 (0.00)
Line_12	97.44 (-11.9%)	91 (-51.1%)	0 (-100.0%)	0 (0.00)
Line_13	96.88 (-11.9%)	74 (-54.6%)	0 (-100.0%)	0 (0.00)
Line_15	91.27 (-13.1%)	24 (-57.9%)	0 (-100.0%)	0 (0.00)
Line_20	67.70 (-17.5%)	0 (-100.0%)	0 (0.0%)	0 (0.00)
Line_21	61.64 (-18.3%)	0	0	0 (0.00)
Line_22	53.39 (-16.5%)	0	0	0 (0.00)

The most substantial optimization consequence can be seen in the *LIH* transformer loading (Figure 23) and in bus magnitudes (Figure 24), this time contained within the respective limits during all time-steps.

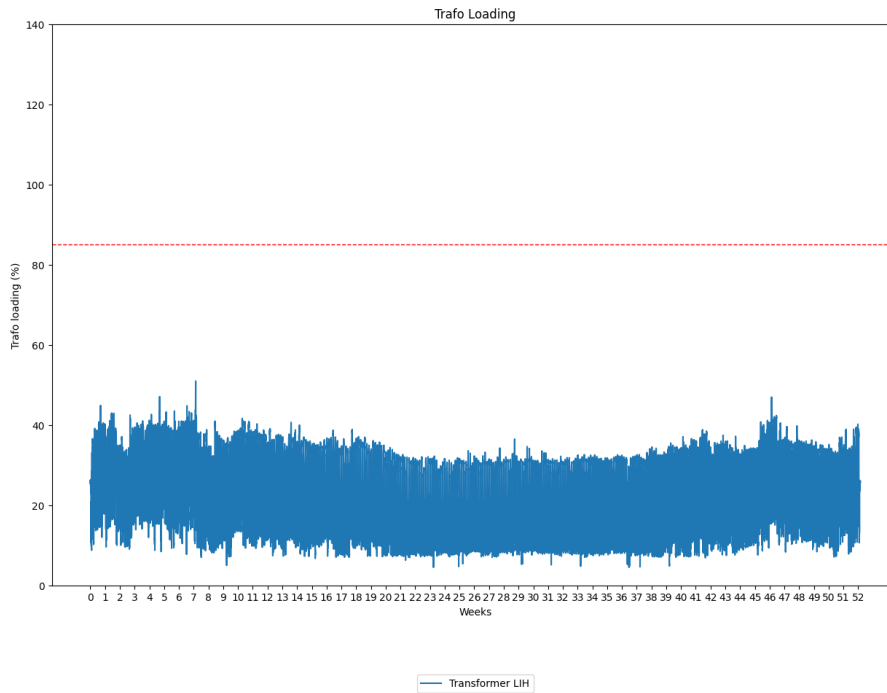
Figure 23: *LIH* transformer loading percentage (optimized EV charging)



Figure 24: Voltage magnitudes of buses closest to the load (optimized EV charging)

5.1.3 PV and BESS integration

The following section evaluates the integration of a PV and battery storage system. The PV energy and the battery dispatch strategy is not only related to the EV demand, but also considers OPS operations and base terminal demand [32]. Specific information about PV and BESS effect on peak, energy and costs can be found in [32]. In this case, all the loads were connected to the MV side of the *LIH* bus. In fact, the BESS receives electricity at 10 kV, and then distributes it to the LV utilities. Consequently, this analysis will not consider the MV/LV *LIH* transformer. The following results are related to Section 4.3 peak minimization model, providing the most relevant insights for this particular study.

Grid Impact

Firstly, analysing the line loading percentages, which revealed to be the most critical metrics in the conventional and optimized scenario, the BESS and PV impact revealed to be relevant, while not completely solve the problem.

In fact, as can be observed in Figure 25, the first lines of the network still exceed the 80% threshold during some time-steps, overstepping also 100% for a minor duration. On the other hand, when looking at lines 20, 21, 22, 23 (Figure 26), which are the

closest to the port's connection point, the stress on them is significantly relieved compared to the optimization scenario results.

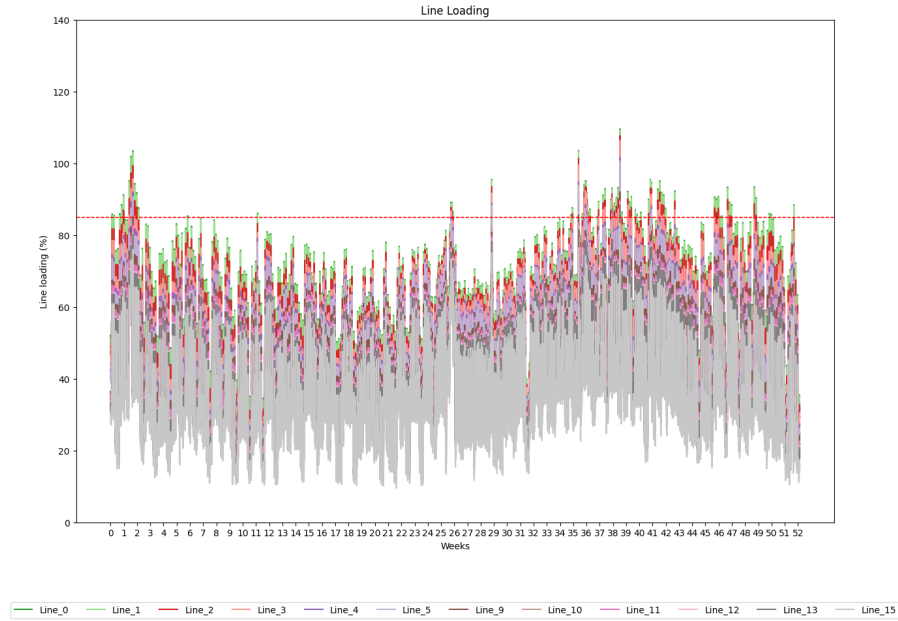


Figure 25: Loading percentage of main transmission corridor lines (BESS and PV integration)

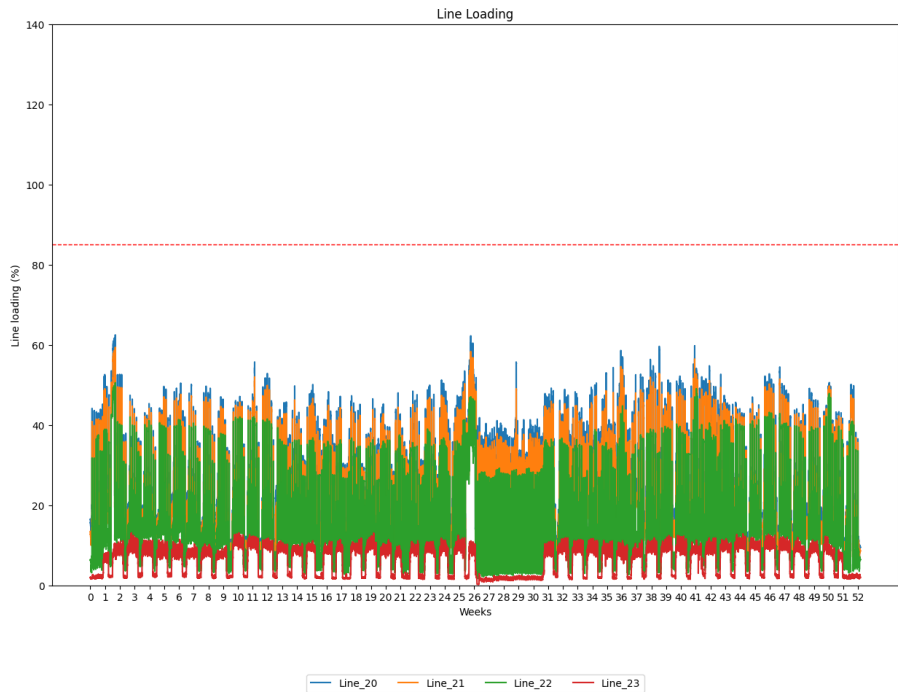


Figure 26: Loading percentage of the lines closest to the load (BESS and PV integration)

To finally evaluate all three scenarios implemented in the Ferry Terminal, Table 11

lists the line loading results, compared to the optimized charging scenario, which already had better performance of the conventional one. Values in brackets indicate the percentage variation with respect to the optimized charging pattern.

Table 11: Line loading statistics under the BESS and PV scenario

Line	Max Loading (%)	15 min Steps $\geq$ 80%	15 min Steps $>$ 100%	Max Consecutive Steps $>$ 100% (h)
Line_0	109.62 (-7.6%)	858 (-46.4%)	30 (-66.7%)	19 (4.75)
Line_1	109.29 (-7.6%)	826 (-46.7%)	30 (-63.9%)	19 (4.75)
Line_2	107.72 (-7.6%)	606 (-50.7%)	20 (-39.4%)	19 (4.75)
Line_3	106.46 (-7.4%)	486 (-49.6%)	19 (-20.8%)	19 (4.75)
Line_4	101.53 (-7.5%)	286 (-59.3%)	8 (-57.9%)	8 (2.00)
Line_5	100.73 (-7.6%)	234 (-60.6%)	5 (-73.7%)	4 (1.00)
Line_9	92.08 (-8.4%)	49 (-80.6%)	0 (-100.0%)	0 (0.00)
Line_10	89.62 (-8.9%)	22 (-84.7%)	0	0 (0.00)
Line_11	89.62 (-8.9%)	22 (-84.7%)	0	0 (0.00)
Line_12	89.07 (-8.6%)	21 (-76.9%)	0	0 (0.00)
Line_13	88.62 (-8.5%)	21 (-71.6%)	0	0 (0.00)
Line_15	83.27 (-8.8%)	16 (-33.3%)	0	0 (0.00)
Line_20	62.53 (-7.6%)	0	0	0 (0.00)
Line_21	59.43 (-3.6%)	0	0	0 (0.00)
Line_22	50.58 (-5.3%)	0	0	0 (0.00)

As can be seen in Table 11, all the main corridor lines faced a maximum loading percentage reduction, with only six of them exceeding the 100% limit, compared to the seven of the precedent scenario. Nevertheless, it can be noted an important reduction in total time-steps over the limit, significantly impacting the stress on the grid, and mainly related to OPS rather than EV charging.

Finally, when analysing the bus voltage magnitudes, also in this case was observed an appreciable stabilization.

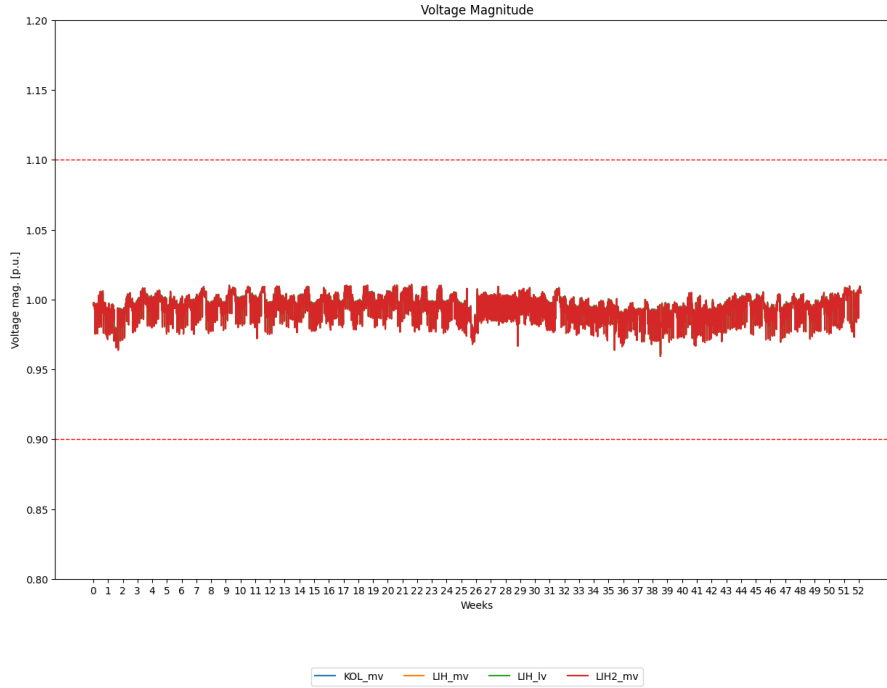


Figure 27: Voltage magnitudes of buses closest to the load (BESS and PV scenario)

5.2 Cargo Terminal

The following section presents Cargo Terminal's simulation results. Due to the uncertainty surrounding the new grid infrastructure that will power the terminal, yet to be planned and constructed, the grid impact assessment is limited to the PV and BESS scenario with optimized EV charging, as this configuration demonstrates the best performance and can better outline network's limitations.

5.2.1 Conventional operation scenario

The conventional scenario operates following the same methodology of the Ferry Terminal but with different vehicles. Particularly, ERG450-65CS (Truck 444 Krokmaskin), as stated in Section 4.2.1, does not follow common rules, but charges whenever idle and is not subject to the intraday skip rule when SOC is higher than 50%.

In this case, as shown in Figure 28, the charging power demand is higher than the Ferry's, due to the different and more numerous vehicles.

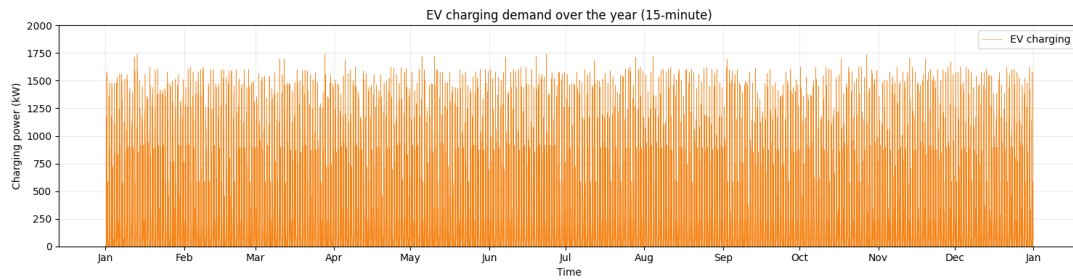


Figure 28: Conventional Cargo Terminal EV demand

It is also insightful to analyse Figure 29, where it can be seen high peaks caused by the uncontrolled charging, sometimes even higher than OPS demand.

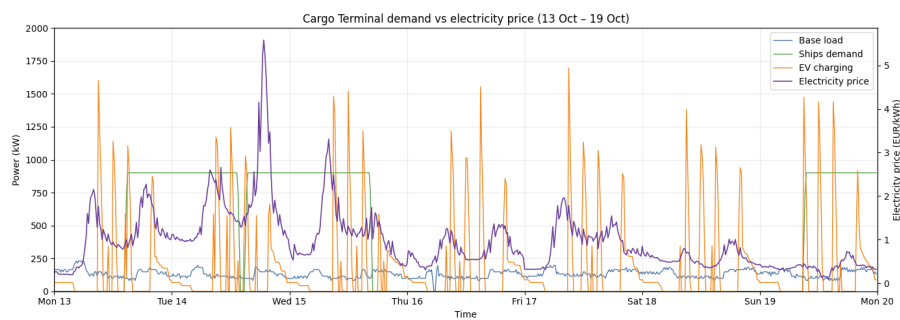


Figure 29: Cargo Terminal demand and electricity prices

Finally, to then evaluate the peak shaving results, Figure 30 and 31 show the sorted energy demand distribution, while Table 12 summarized most relevant KPI.

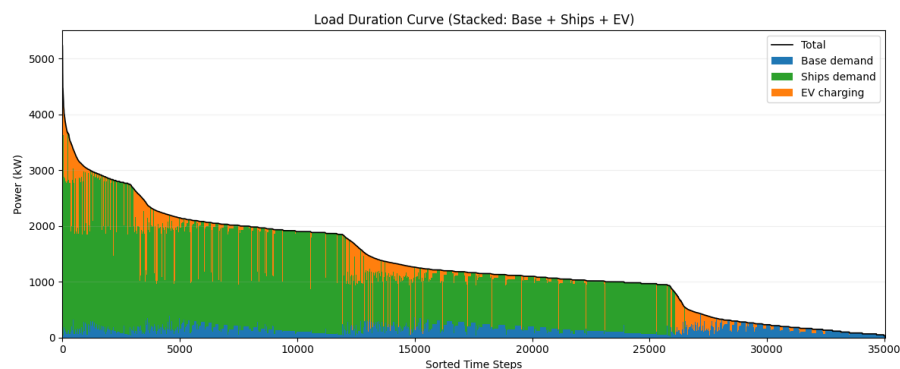


Figure 30: Cargo Terminal sorted energy demand (conventional EV charging)

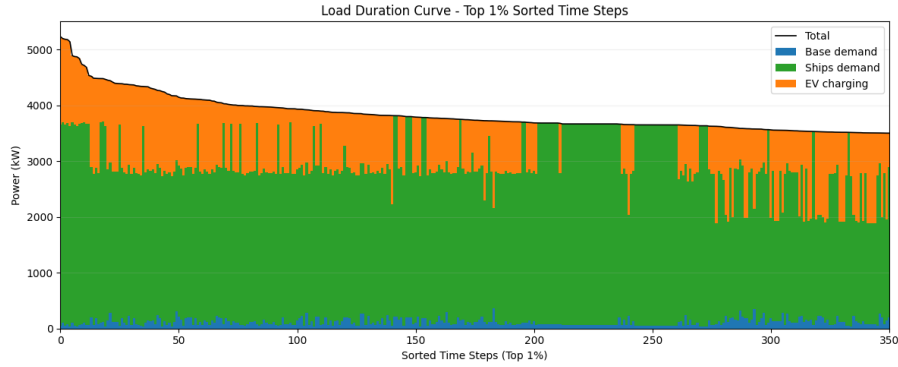


Figure 31: Cargo Terminal top 1% sorted energy demand (conventional EV charging)

Table 12: Conventional EV charging results

Metric	Value
Peak demand without charging	3.81 MW
Peak demand with charging	5.23 MW
Peak demand increase	1414.46 kW
EV charging demand at system peak	1579.0 kW
Total charged energy	1,548,698 kWh
Total charging cost	879,676 SEK

5.2.2 Optimized charging scenario

As for the Ferry Terminal but with different vehicles, this configuration follows the framework illustrated in Section 4.2.2. Specifically, ERG450-65CS (Truck 444 Krokmaskin) is optimized in a secondary separate model, using as inputs the other EV optimization results. As can be seen comparing Figure 32 and 33 with Figure 28, it was possible to reduce peaks during several time-steps.

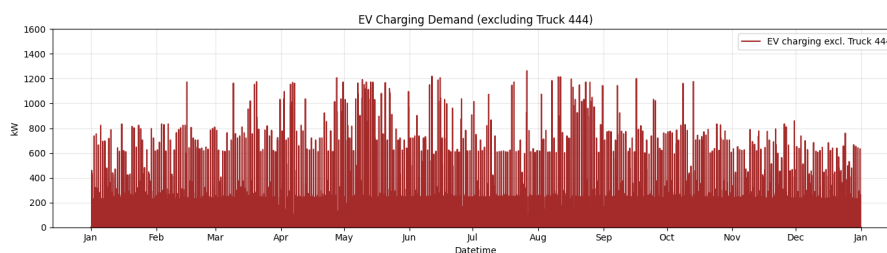


Figure 32: Optimized Cargo Terminal EV demand (excluding Truck 444)

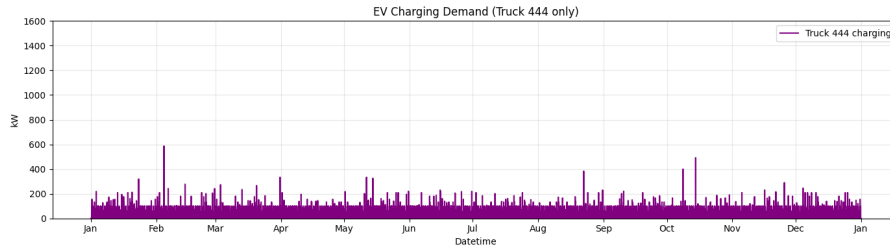


Figure 33: Truck 444 optimized demand

The simultaneous presence of multiple vessels with high and fast power demand, limits the ability of achieving consistent peak shaving throughout the year, with some charging peaks remaining visible in the optimized profile. Nevertheless, Truck 444 Krokmaskin had a significant impact on the overall charging demand, reaching power peaks of approximately 600 kW. Due to its intensive energy consumption and operational schedule, its charging demand cannot be distributed as evenly as that of the other vehicles, resulting in concentrated charging periods that contribute substantially to the total load. However, these peaks generally occur during periods when vessel demand is low or absent, allowing the available network capacity to be utilized for EV charging without significantly increasing the overall system peak.

In Figure 34 is shown the charging pattern compared to the electricity prices, showing the notable difference from the conventional scenario.

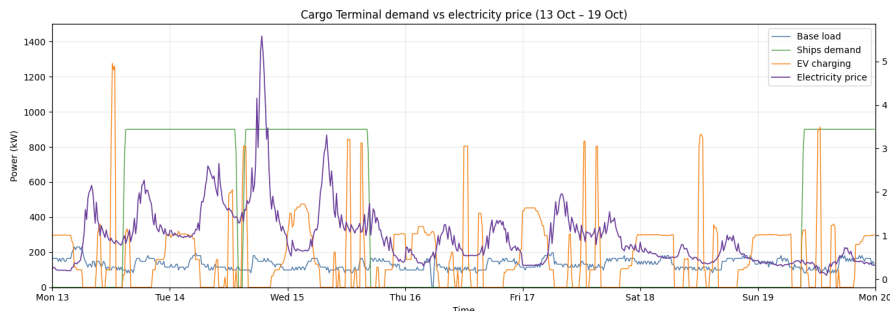


Figure 34: Cargo Terminal demand and electricity prices

Finally, Figure 35, 36 plot the sorted time-steps, while Table 13 compares the conventional charging results with the optimized ones.



Figure 35: Cargo Terminal sorted energy demand (optimized EV charging)

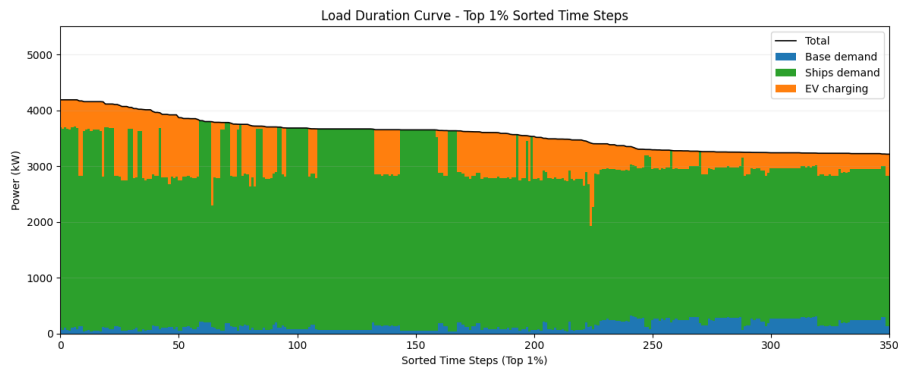


Figure 36: Cargo Terminal top 1% sorted energy demand (optimized EV charging)

When evaluating Table 13, particularly interesting is Truck 444 contribution. In fact, its singular impact at the system peak is of 132.95 kW (roughly 35% of the total increase); its charged energy is 335272 kWh (roughly 28% of the total), with consequent charging costs of 129454.44 SEK (roughly 20% of the total), resulting in a highly significant contribution among the 20 optimized vehicles.

Table 13: Comparison between conventional and optimized EV charging scenarios

Metric	Conventional	Optimized	Variation (%)
Peak demand without charging	3.81 MW		N.A.
Peak demand with charging	5.23 MW	4.19 MW	-19.9%
Peak demand increase	1414.46 kW	373.7 kW	-73.6%
EV charging demand at system peak	1579.0 kW	488.9 kW	-69.0%
Total charged energy	1,548,698 kWh	1,546,221 kWh	+0.1%
Total charging cost	879,676 SEK	640,403 SEK	-27.2%

5.2.3 PV and BESS integration

The following section evaluates the integration of a PV and battery storage system. The methodology is the same as Section 5.1.3: all the loads are connected to the MV side of the cargo's connection point bus (KLU_{new}), where the BESS receives electricity at 10 kV and then distributes it to the utilities. This simulation also comprehends the new Ferry Terminal load integrating PV and BESS, since the Cargo Terminal will be the second one to be constructed.

Grid impact

Firstly analysing the line loading percentages, it is immediate noticing how the EV and OPS implementation of the new Cargo Terminal has an important impact on the network, this time reaching peaks of roughly 147%.

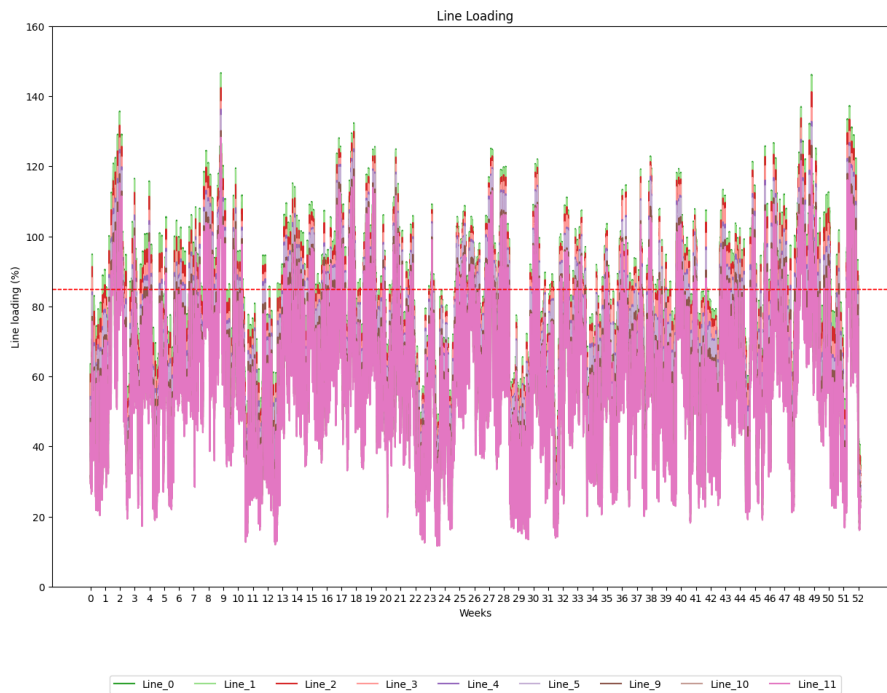


Figure 37: Loading percentage of main transmission corridor lines

As stated in Table 14, all the lines before the connection point (KLU_{new} bus), so from Line 0 to Line 11, reach values over 128% and maintain overloading status for more than 20 consecutive hours, with peaks of 36 hours, which is infeasible to be supported by the grid.

Table 14: Line loading statistics with both Cargo and Ferry Terminals with PV and BESS

Line	Max Loading (%)	15m steps $\geq 80\%$	15m steps $> 100\%$	Max Consecutive steps $> 100\%$ (h)
Line_0	146.70	10279	2364	144 (36.00)
Line_1	146.32	10092	2258	144 (36.00)
Line_2	142.43	8856	1721	128 (32.00)
Line_3	138.52	7675	1312	112 (28.00)
Line_4	136.30	6671	958	101 (25.25)
Line_5	134.55	6185	854	100 (25.00)
Line_9	130.15	4412	515	99 (24.75)
Line_10	128.25	3800	422	80 (20.00)
Line_11	128.25	3800	422	80 (20.00)
Line_12	68.58	0	0	0 (0.00)
Line_13	68.18	0	0	0 (0.00)
Line_15	64.02	0	0	0 (0.00)
Line_20	54.24	0	0	0 (0.00)
Line_21	51.13	0	0	0 (0.00)
Line_22	50.85	0	0	0 (0.00)

Finally, when evaluating the most relevant buses voltage magnitudes, it can be observed that during all time-steps, values are among the threshold.

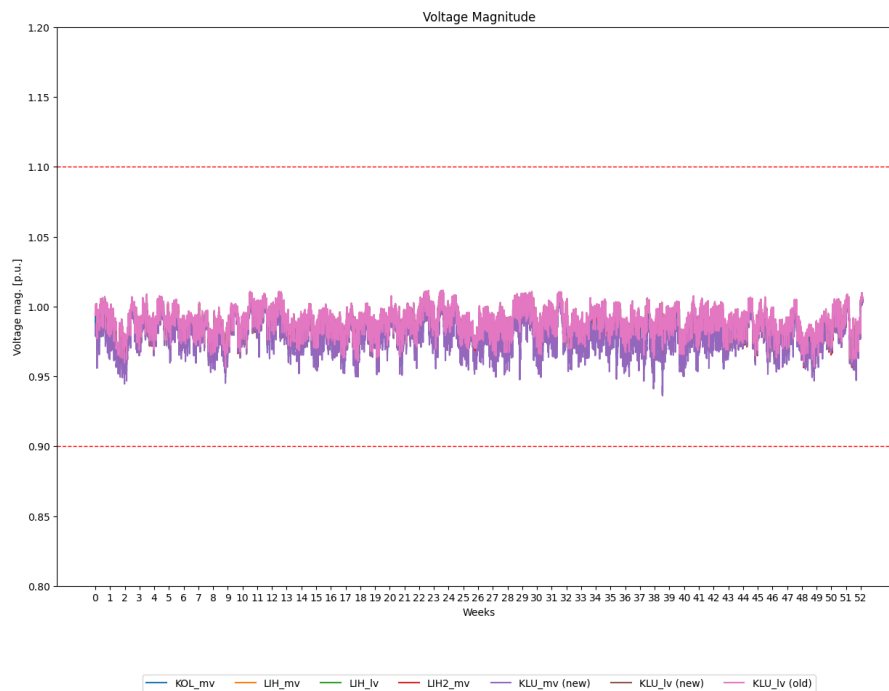


Figure 38: Voltage magnitudes of buses closest to the Ferry and Cargo loads

Chapter 6

Discussion

Results show how the electrification of Oskarshamn's port operations can significantly impact the local distribution network, particularly when OPS demand and industrial electric vehicles are simultaneously integrated into the system. To isolate the impact of the new terminals' electrification, the grid impact assessments were performed without considering OPS of the existing terminals. This specific study can be found in Appendix C. Consequently, the obtained results can be evaluated as the incremental impact associated with the planned electrification projects.

The Ferry Terminal study demonstrated that the proposed optimization framework can significantly mitigate the electrification's effects on the network. In fact, the combination of optimized EV charging, PV generation, and BESS, resulted in a substantial reduction in line and transformer loadings, while also stabilizing buses voltage magnitudes. With the PV and BESS integrated simulation, only 30 time-steps exceeded the 100% loading threshold during the year, corresponding to approximately 0.09% of the annual time-steps. This represents a reduction compared to the conventional scenario, which experienced a total of 158 overload periods (0.5% of the year). Considering the conservative assumptions adopted, the results suggest that the network can operate safely for almost the entire year without requiring major reinforcements. Nevertheless, its limited operational margin indicates that concerns regarding network fragility should not be overlooked.

The situation is considerably different when the Cargo Terminal is included in the analysis. In this case, overload conditions are more frequent (2,364 time-steps – 7% of the year), with maximum continuous periods of 36 hours and loading levels reaching

147%. These results highlight the impact of the Cargo Terminal, supporting the coherent decision to connect the facility to a new section of the network (Oskarshamn Energi).

More generally, the outcomes emphasize that optimization techniques alone cannot completely solve all network criticalities. Nevertheless, substantial improvements can be achieved, despite the framework was only applied to port-related loads. This suggests that if other industrial and commercial consumers would adopt similar strategies, the network could better support electrified utilities integration.

It should also be considered that the achieved outcomes are the result of a model involving multiple stakeholders with different objectives: primarily Smålandshamn AB, which operates the port infrastructure, and Oskarshamn Energi, which operates the electrical network. In fact, the proposed methodology aimed to reduce network stress and improving asset utilization, potentially deferring grid reinforcement investments, while also lowering the Port's electricity procurements costs and maintaining operationally feasible charging schedules. If a purely grid-oriented perspective had been adopted, further improvements could likely have been achieved, resulting in lower line loadings, fewer overload occurrences, and a more pronounced peak reduction. However, providing direct economic benefits to both parties increases the practical attractiveness and likelihood of adopting such a solution.

An important observation throughout all simulated scenarios is that the most critical network elements are those located in the first branch of the feeder, supplying downstream loads. The analysis therefore suggests that future planning efforts should primarily focus on increasing the capacity of this upstream corridor rather than on the terminal connection points themselves.

It should be also noted that the simulations were performed using historical data, which may progressively change over the upcoming years. Future developments in the surrounding area, such as increased PV installations or wider adoption of district-heating solutions instead of HVAC system, could reduce local electricity consumptions. At the same time, the opposite effect could arise from further electrification of industrial utilities, highlighting the importance of continuous grid monitoring and forecasting.

Nevertheless, the port's electrification, as considered within the methodology adopted in this study, can completely eliminate the port's direct CO_2 emissions, with important

benefit for the surrounding population, as emissions associated with berthed vessels and fossil-fuelled port equipment directly affects air quality in nearby areas.

Finally, results indicate that smart charging together with local renewable generation and battery storage systems represent effective tools to mitigate the impact of a large facility as the Port of Oskarshamn, particularly when considering OPS demand and following EU electrification directives. While these solutions may not completely eliminate grid reinforcement, they can significantly reduce network stress, defer infrastructure investments and improve the integration of new electrical loads.

Chapter 7

Conclusions

The thesis modelled and evaluated the planned electrification of Oskarshamn port's new Ferry and Cargo Terminals. A Mixed-Integer Linear Programming (MILP) optimization tool was designed and used to model EV charging, integrating it into a complex scenario that also included passenger and cargo vessels' onshore power supply (OPS). Moreover, renewable energy integration and power flow analysis were performed to assess the system's impact on the electrical grid.

The results demonstrated that uncontrolled electrification would lead to a significant increase in peak demand, creating critical operating conditions within the network. The Ferry Terminal's proposed charging optimization eliminated the contribution of vehicles' electrical demand to the annual system peak, while also reducing charging costs by approximately 18% and decreasing maximum line loading by roughly 9–18% across the most critical lines. In addition, transformer overloading was successfully eliminated and bus voltage magnitudes stabilized.

The integration of PV generation and BESS provided additional benefits by further reducing line loading by 7–9%, while decreasing the number of overload occurrences. The results showed how the combination of optimized charging with local energy resources and battery storage can effectively mitigate the electrification's impact while maintaining operational feasibility.

The Cargo Terminal presented a more challenging scenario due to the higher OPS demand and the presence of numerous energy-intensive cargo handling vehicles. Nevertheless, the optimization framework was able to reduce the EV related peak demand by roughly 70% and reducing charging costs by approximately 27%.

Overall, the study identified the first feeder branch as the most critical one, indicating that future reinforcement efforts should be focused on those lines instead of the downstream corridors. The results also showed the flexibility measures' value, with PV and BESS integration progressively reducing grid impact.

Finally, findings revealed that grid reliability, economic efficiency, and EU policies compliance do not necessarily conflict among each other. Through intelligent flexibility management, it is possible to simultaneously reduce operational costs for the Port and technical constraints for the DSO, supporting sustainable port electrification while creating benefits for both stakeholders and contributing to the reduction of CO_2 emissions.

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Appendix A

EV and charging optimization model

A.1 Working schedule generator

Each EV schedule is different and was randomly generated day-by-day, to create a variable and realistic pattern. The schedule generator was fundamental to determine each vehicle's consumptions in different part of the day, optimally charging it especially during day-time allowed charging windows.

Fixed parameters of each vehicle are listed in Table A.1 and obtained from [35]. Battery capacity (kWh), average discharge rate (kWh/h) and fast charge capacity (kW) values were already presented in Table 1.

Table A.1: Fleet vehicle operational and charging parameters

ID	Category	Model		Total Working Hours	Working Duration Capacity (h)	Requires Charging During Operation
Truck 417	Forklift	Merlinum 6-600	EA	2	4.6	no
Truck 405	Forklift	Merlinum 6-600	EA	8	4.6	yes
Truck 413	Forklift	Merlinum 6-600	EA	4	4.6	no
Truck 402	Forklift	Merlinum 6-600	EA	2	4.6	no
Truck 420	Forklift	SMV 16 Ever		11	5.5	yes
Truck 422	Forklift	SMV 16 Ever		11	5.5	yes

ID	Category	Model	Total Working Hours	Working Duration Capacity (h)	Requires Charging During Operation
Truck 423	Forklift	SMV 16 Ever	11	5.5	yes
Truck 424	Forklift	SMV 16 Ever	11	5.5	yes
Truck 425	Forklift	SMV 16 Ever	11	5.5	yes
Truck 426	Forklift	SMV 16 Ever	11	5.5	yes
Truck 427	Forklift	SMV 16 Ever	11	5.5	yes
Truck 428	Forklift	SMV 16 Ever	11	5.5	yes
Truck 429	Forklift	SMV 16 Ever	11	5.5	yes
Truck 421	Forklift	SMV 16 Ever	11	5.5	yes
Lastmaskin 430	Loader	Volvo L120H	4	5.0	no
Lastmaskin 431	Loader	Volvo L120H	4	5.0	no
Lastmaskin 432	Loader	Volvo L120H	4	5.0	no
Lastmaskin 435	Loader	Volvo L120H	4	5.0	no
Lastmaskin 437	Loader	Volvo L120H	8	5.0	yes
Lastmaskin 439	Loader	Volvo L120H	2	5.0	no
Lastmaskin 438	Loader	Volvo L120H	11	5.0	yes
Lastmaskin 433	Loader	Volvo L120H	11	5.0	yes
Lastmaskin 436	Loader	Volvo L120H	3	5.0	no
Truck 442	Reach Stacker	SMV 46/32 E-ver	4	5.5	no
Truck 443	Reach Stacker	SMV 46/32 E-ver	4	5.5	no
Truck 444 Krokmaskin	Reach Stacker	ERG450-65CS	8	5.1	yes
Term.drag 468	Terminal Mover	Terberg RT 223 EL.	4	6.7	no
Term.drag 469	Terminal Mover	Terberg RT 223 EL.	4	6.7	no
Term.drag 470	Terminal Mover	Terberg RT 223 EL.	4	6.7	no
Term.drag 471	Terminal Mover	Terberg RT 223 EL.	4	6.7	no
Term.drag 472	Terminal Mover	Terberg RT 223 EL.	4	6.7	no
Term.drag 475	Terminal Mover	Terberg RT 223 EL.	4	6.7	no

A.2 Optimization framework

This section describes how EV charging is optimised at the ferry and cargo terminals. Both cases use the same basic idea: find a charging schedule that respects vehicle operation, limits terminal load, produces a smooth charging pattern, and reduces electricity cost. The problem is solved as a *lexicographic* mixed-integer linear program

(MILP), meaning the objectives are optimised one after another in a fixed priority order, rather than combined into a single weighted sum.

A.2.1 Common setup

Time and vehicles. Both models use 15-minute timesteps over one full year:

$$\Delta t = 0.25 \text{ h}, \quad |\mathcal{I}| = 35,040.$$

Let $\mathcal{V}$ be the set of vehicles ($|\mathcal{V}| = 32$) and $\mathcal{I} = \{0, \dots, |\mathcal{I}| - 1\}$ the set of timesteps.

Main decision variables. For each vehicle $v \in \mathcal{V}$ and timestep $i \in \mathcal{I}$:

- $P_{v,i}$ [kW]: charging power,
- $\text{SOC}_{v,i}$ [kWh]: battery state of charge.

At fleet level, the model also uses:

- P^{peak} [kW]: maximum terminal load,
- $y_{v,i}^{\text{fast}} \in \{0, 1\}$: whether fast charging is active for eligible vehicles.

Known inputs. The following quantities are treated as fixed inputs:

- $E_{v,i}^{\text{cons}}$ [kWh]: vehicle energy use in interval i ,
- Cap_v [kWh]: battery capacity,
- P_i^{base} [kW]: terminal base load,
- P_i^{shore} [kW]: on-shore power demand from ship operations,
- π_i [currency/kWh]: day-ahead electricity price.

The fixed terminal load is

$$D_i = P_i^{\text{base}} + P_i^{\text{shore}}.$$

A ship-presence flag is also used:

$$s_i = \begin{cases} 1, & \text{if } P_i^{\text{shore}} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Aggregate charging power. Total EV charging at timestep i is

$$P_i^{\text{tot}} = \sum_{v \in \mathcal{V}} P_{v,i}.$$

Important modelling assumption. Charging efficiency is not included. The model assumes that 1 kWh taken from the grid adds 1 kWh to the battery.

A.2.2 Battery and operating constraints

Battery levels are limited to

$$0.1 \text{ Cap}_v \leq \text{SOC}_{v,i} \leq \text{Cap}_v.$$

At the start of the year, all vehicles begin fully charged:

$$\text{SOC}_{v,0} = \text{Cap}_v.$$

During the year, each vehicle must again be fully charged at every daily 07:00 checkpoint.

The battery balance is

$$\text{SOC}_{v,i+1} = \text{SOC}_{v,i} + P_{v,i} \Delta t - E_{v,i}^{\text{cons}}. \quad (\text{A.1})$$

The model also enforces the following operating rules:

- no charging while a vehicle is consuming energy,
- no charging above available battery capacity,
- charging only in allowed time windows in the fleet model,
- slow charging limited to 22 kW,
- fast charging limited by vehicle-specific maximum power for eligible vehicles.

The terminal load, including EV charging, must not exceed the peak variable:

$$D_i + P_i^{\text{tot}} \leq P^{\text{peak}} \quad \forall i \in \mathcal{I}. \quad (\text{A.2})$$

A.2.3 Lexicographic optimisation

The model is solved in four sequential stages:

$$\min P^{\text{peak}} \rightarrow \min E^{\text{tot}} \rightarrow \min J^{\text{smooth}} \rightarrow \min C^{\text{charge}}.$$

After each stage, the result of that stage is kept fixed within a small tolerance before moving to the next one.

Stage 1: minimise peak demand. The first objective is to reduce the maximum terminal load. This is the main grid-related goal.

Stage 2: minimise total charged energy. Among all solutions with the minimum peak from Stage 1, Stage 2 selects the one with the lowest total charged energy:

$$E^{\text{tot}} = \sum_{i \in \mathcal{I}} \sum_{v \in \mathcal{V}} P_{v,i} \Delta t.$$

This stage acts as a tie-breaker: it removes unnecessary charging while keeping the best peak.

Stage 3: improve charging shape. Stage 3 makes the charging profile smoother and more operationally acceptable. It penalises:

- upward changes in EV charging power,
- upward changes in terminal load,
- charging during ship presence,
- charging power above 100 kW,
- deviation from a reference charging profile.

The reference profile is built after Stage 2 as a mostly flat charging level, with a small adjustment toward timesteps where more base-load capacity is available.

The smoothness objective is

$$J^{\text{smooth}} = 8 \sum_i (r_i^{\text{ev}} + r_i^{\text{term}} + d_i) + 20 \sum_i s_i P_i^{\text{tot}} + 30 \sum_i e_i, \quad (\text{A.3})$$

where the auxiliary variables are defined by

$$r_i^{\text{ev}} \geq P_{i+1}^{\text{tot}} - P_i^{\text{tot}}, \quad (\text{A.4})$$

$$r_i^{\text{term}} \geq L_{i+1} - L_i, \quad (\text{A.5})$$

$$e_i \geq P_i^{\text{tot}} - 100, \quad (\text{A.6})$$

$$d_i \geq \left| P_i^{\text{tot}} - P_i^{\text{ref}} \right| \quad \text{in allowed charging periods.} \quad (\text{A.7})$$

Only *upward* ramps are penalised.

Stage 4: minimise electricity cost. Finally, the model shifts charging toward lower-price periods while keeping the earlier results nearly unchanged:

$$C^{\text{charge}} = \sum_{i \in \mathcal{I}} \sum_{v \in \mathcal{V}} P_{v,i} \Delta t \pi_i.$$

Post-processing cost. For reporting, network charges not included in π_i are added afterwards:

$$C^{\text{report}} = 1.0561 C^{\text{charge}} + 0.025 E^{\text{tot}}.$$

A.2.4 Ferry terminal implementation

At the ferry terminal, all vehicles are optimised together in one model.

The terminal ramp term uses only base load plus EV charging:

$$L_i = P_i^{\text{base}} + P_i^{\text{tot}}.$$

Shore power is included in the peak constraint (A.2), but not in the ramp penalty.

Fleet charging rules are:

- slow charging: 20:00–06:45,
- fast charging: 09:00–10:45, 12:00–13:45, 15:00–16:45,
- at most 4 simultaneous fast chargers,
- maximum upward EV ramp: 20 kW per 15-minute step.

A.2.5 Cargo terminal implementation

The cargo terminal uses the same framework, but in two steps.

Step 1: fleet optimisation. First, all vehicles except Truck 444 Krokmaskin are optimised together using the same four-stage procedure. Truck 444 is excluded from the fleet consumption dataset in this step and is handled separately afterwards.

The fleet model is the same as at the ferry terminal, except for:

- cargo terminal base load and cargo ship schedule,
- at most 5 simultaneous fast chargers,
- maximum upward EV ramp: $|\mathcal{V}| \times 22 = 704$ kW.

Step 2: dedicated Truck 444 optimisation. After the fleet solution is known, Truck 444 is optimised alone in a second four-stage MILP. The charging of all other vehicles is fixed, and the background load becomes

$$B_i = D_i + \sum_{v \neq 444} P_{v,i}^*.$$

The peak constraint is then

$$B_i + P_{444,i} \leq P^{\text{peak}}.$$

Truck 444 follows simpler charging rules:

- it may charge whenever it is not working,
- it is not restricted to slow/fast clock windows,
- no fast-charging binary variables are used,
- charging is limited by the truck maximum power,
- the ramp penalty uses $L_i = B_i + P_{444,i}$.

Because Truck 444 is re-optimised on top of the fleet result, the final cargo-terminal peak may be higher than the fleet-only optimum.

The final cargo-terminal result combines:

- fleet charging for all vehicles except Truck 444,

- the dedicated Truck 444 charging profile,
- post-processed costs from both optimisations.

A.2.6 Summary

In both terminals, the optimisation framework has the same structure:

1. respect vehicle battery and operating constraints,
2. minimise terminal peak demand,
3. remove unnecessary charging,
4. smooth the charging profile,
5. reduce electricity cost as far as possible without breaking the earlier priorities.

The main difference is that the ferry terminal solves one joint model for the full fleet, whereas the cargo terminal solves a fleet model first and then a separate model for Truck 444 with relaxed charging rules.

Appendix B

Electrical network parameters

B.1 Lines

Line connections are shown in Table B.2. The values of resistance (Ω/km), reactance (Ω/km), capacitance (nF/km), and maximum current (kA) are assumed equal for all lines and set to $0.21 \Omega/\text{km}$, $0.09 \Omega/\text{km}$, $0.35 \text{ nF}/\text{km}$, and 0.265 kA , respectively. Values were taken from [35] and validated by Oskarshamn Energi.

Table B.2: Distribution network line data

Name	From	To	Length (km)	Diameter (m)
Line_0	Bus_0	Bus_2	0.375	0.285
Line_1	Bus_2	Bus_4	0.081	0.285
Line_2	Bus_4	Bus_6	0.334	0.450
Line_3	Bus_6	Bus_8	0.660	0.450
Line_4	Bus_8	Bus_13	1.011	0.450
Line_5	Bus_13	Bus_19	0.393	0.285
Line_6	Bus_19	Bus_23	0.412	0.450
Line_7	Bus_23	Bus_25	0.143	0.720
Line_8	Bus_23	Bus_57	0.497	0.720
Line_9	Bus_19	Bus_28	0.297	0.450
Line_10	Bus_28	Bus_30	0.288	0.450
Line_11	Bus_30	Bus_33	0.185	0.450
Line_12	Bus_33	Bus_36	0.558	0.285
Line_13	Bus_36	Bus_38	0.315	0.285
Line_14	Bus_38	Bus_40	0.441	0.450
Line_15	Bus_38	Bus_44	0.605	0.450
Line_16	Bus_44	Bus_46	0.729	0.285
Line_17	Bus_46	Bus_48	0.417	0.450
Line_18	Bus_48	Bus_50	0.473	0.285
Line_19	Bus_50	Bus_51	0.173	0.450
Line_20	Bus_44	Bus_53	0.398	0.285
Line_21	Bus_53	Bus_55	0.366	0.450
Line_22	Bus_55	LIH_mv	0.180	0.450
Line_23	LIH_mv	LIH2_mv	0.482	0.450

B.2 Transformers

Table B.3 lists all the transformers in the evaluated side of the network. Specifically, KLU_{old} and KLU_{new} are related to the existing and new Cargo terminals, while LIH is linked to the new Ferry Terminal and LIH_2 to a candle factory. EXT is the transformer stepping down the 50 kV electricity from E.ON to 10 kV city distribution level.

Table B.3: Transformer parameters of the distribution network

Name	From	To	Rated Power (MVA)	HV Voltage (kV)	LV Voltage (kV)	Short-circuit Voltage (%)	Resistive Voltage Drop (%)	Iron Losses (kW)	No-load Current (%)
EXT	EG	Bus_0	25	50	10	4	0.3	30	0.1
OGP	Bus_2	Bus_3	0.5	10	0.4	12	0.41	14	0.07
TBD	Bus_4	Bus_5	0.5	10	0.4	12	0.41	14	0.07
URA	Bus_6	Bus_7	0.5	10	0.4	12	0.41	14	0.07
SVN	Bus_8	Bus_9	0.8	10	0.4	12	0.41	14	0.07
LAX	Bus_13	Bus_14	1.6	10	0.4	12	0.41	14	0.07
KBV	Bus_19	Bus_20	0.8	10	0.4	12	0.41	14	0.07
NSG	Bus_23	Bus_24	0.8	10	0.4	12	0.41	14	0.07
KRVA	Bus_25	Bus_26	0.8	10	0.4	12	0.41	14	0.07
JUN1	Bus_57	Bus_58	0.8	10	0.4	12	0.41	14	0.07
JUN2	Bus_57	Bus_58	0.8	10	0.4	12	0.41	14	0.07
JUN3	Bus_57	Bus_58	1.0	10	0.4	12	0.41	14	0.07
JUN4	Bus_57	Bus_58	0.8	10	0.4	12	0.41	14	0.07
JUN5	Bus_57	Bus_58	1.0	10	0.4	12	0.41	14	0.07
JUN6	Bus_57	Bus_58	1.0	10	0.4	12	0.41	14	0.07
JUN7	Bus_57	Bus_58	0.8	10	0.4	12	0.41	14	0.07
JUN8	Bus_57	Bus_58	0.8	10	0.4	12	0.41	14	0.07
JUN9	Bus_57	Bus_58	1.0	10	0.4	12	0.41	14	0.07
JUN10	Bus_57	Bus_58	1.0	10	0.4	12	0.41	14	0.07
JUN11	Bus_57	Bus_58	0.5	10	0.4	12	0.41	14	0.07
BVH	Bus_28	Bus_29	0.8	10	0.4	12	0.41	14	0.07
KLU_old	Bus_33	Bus_34	0.8	10	0.4	12	0.41	14	0.07
KLU_new	Bus_33	Bus_35	0.8	10	0.4	12	0.41	14	0.07
SPL	Bus_36	Bus_37	0.8	10	0.4	12	0.41	14	0.07
OLJ	Bus_38	Bus_39	0.8	10	0.4	12	0.41	14	0.07
HDM	Bus_40	Bus_41	0.8	10	0.4	12	0.41	14	0.07
KAP	Bus_44	Bus_45	0.8	10	0.4	12	0.41	14	0.07
HUK	Bus_46	Bus_47	0.8	10	0.4	12	0.41	14	0.07
LAN	Bus_48	Bus_49	0.8	10	0.4	12	0.41	14	0.07
MAN	Bus_51	Bus_52	0.5	10	0.4	12	0.41	14	0.07
KSO	Bus_53	Bus_54	0.8	10	0.4	12	0.41	14	0.07
KOL	Bus_55	Bus_56	0.8	10	0.4	12	0.41	14	0.07
LIH	LIH_mv	LIH_lv	0.8	10	0.4	12	0.41	14	0.07
LIH2	LIH2_mv	LIH2_lv	0.8	10	0.4	12	0.41	14	0.07

Appendix C

Additional simulations

In the following chapter, additional network simulations are presented, with the goal of giving a complete understanding of the results presented.

C.1 Grid analysis with no additional Port loads

The following table presents the simulation's results representing the actual network situation, with no new Ferry and Cargo Terminals loads considered. As can be seen, lines particularly of the main corridor, already show high maximum loading percentages, with peaks of roughly 78%.

Table C.4: Line loading statistics

Line	Max Loading (%)	15m steps $\geq 80\%$	15m steps $> 100\%$
Line_0	77.95	0	0
Line_1	77.62	0	0
Line_2	76.02	0	0
Line_3	74.76	0	0
Line_4	69.86	0	0
Line_5	69.07	0	0
Line_9	60.50	0	0
Line_10	58.06	0	0
Line_11	58.06	0	0
Line_12	57.51	0	0
Line_13	57.06	0	0
Line_15	51.79	0	0
Line_20	30.01	0	0
Line_21	24.38	0	0
Line_22	13.43	0	0

C.2 OPS in both existing and new Ferry Terminals

Once the new Ferry Terminal is going to be built and its utilities electrified, the existing Ferry Terminal operations will be uncertain. To simulate the grid impact of both Ferry Terminals simultaneous operations, the following assumption related to the existing Ferry Terminal have been made:

- Electrified demand consists of the base load and OPS operations, with no EV been included.
- After the new terminal opens, the existing terminal retains only 30% of arrivals (70% reduction), assuming that both terminals are unlikely to run at full ship volume simultaneously. Maintaining the same OPS dimensioning method used along the report, stays are then sampled so each ship keeps about the same share of traffic as in the original schedule (136 calls of which roughly half are DROTEN and a quarter each VISBY and GOTLAND). Furthermore, each kept stay keeps its

duration, but its date and time of day are moved randomly across the year. Stays are placed so only one ship is at the existing terminal at a time; if two would overlap, the later arrival is shifted forward in 15-minute steps until it fits.

- Base load is scaled down by a 30%.
- Existing Ferry Terminal is connected to OLJ bus, in medium voltage.

The results are shown in Figures 39 and 40, showing the main corridor line loading percentages and bus voltage magnitudes, respectively.

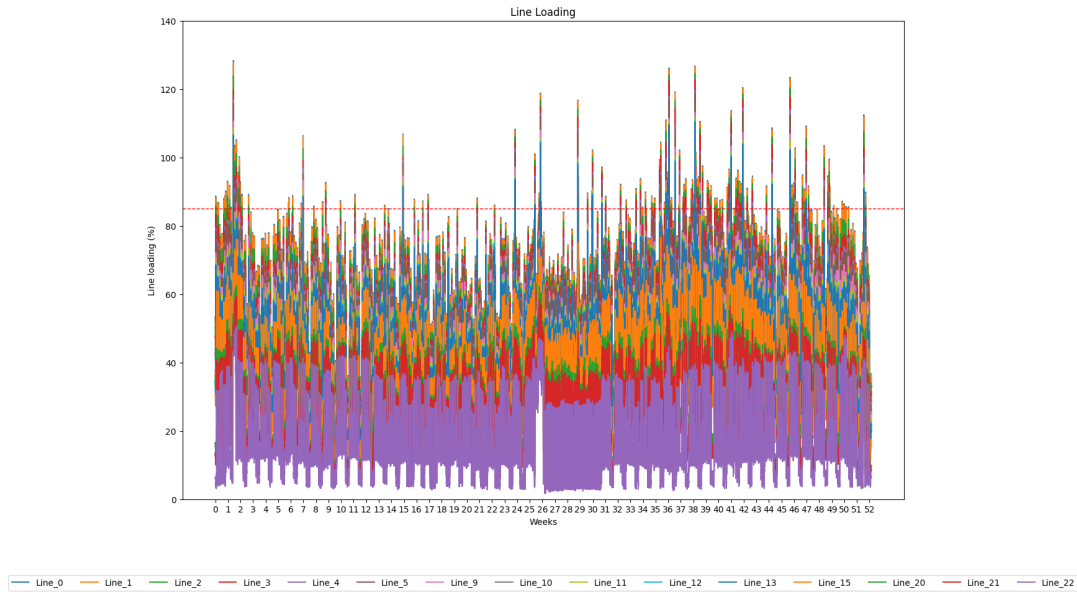


Figure 39: Loading percentage of main transmission corridor lines

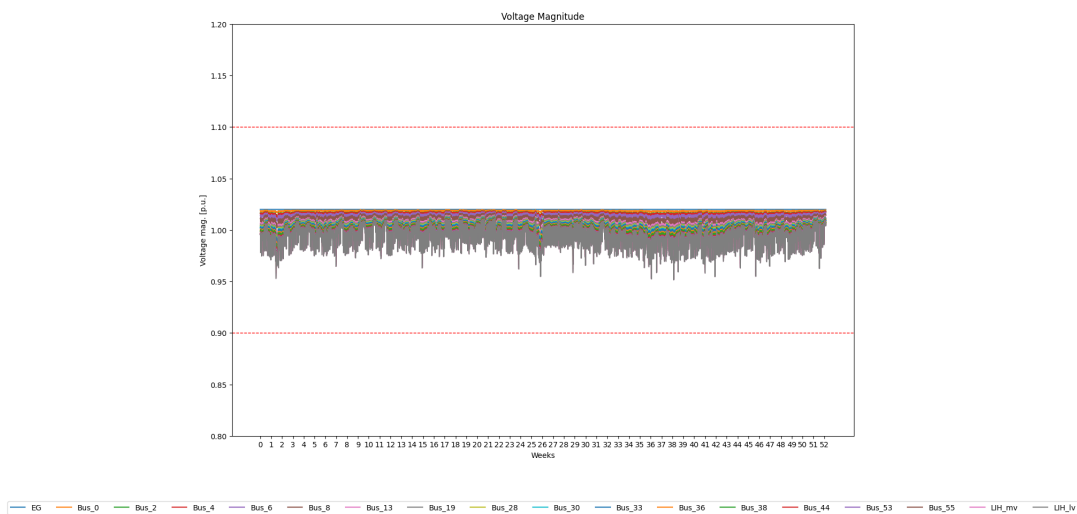


Figure 40: Voltage magnitudes of buses closest to the new and existing Ferry loads

Moreover, Table C.5, representing only critical lines, illustrates the comparison

between the New Ferry Terminal electrification following the peak minimization EV, PV and BESS strategy, with the scenario including OPS operations also in the existing terminal.

Table C.5: Line loading results under the considered terminal electrification scenarios

Line	Ferry T. PV + BESS (peak min.)		Existing and New Ferry Terminals	
	Max loading (%)	Max consecutive hours over 100%	Max loading (%)	Max consecutive hours > 100%
0	109.62	4.75	128.38	4.75
1	109.29	4.75	127.93	4.75
2	107.72	4.75	124.61	4.75
3	106.46	4.75	122.75	4.75
4	101.53	2	119.62	3
5	100.73	1	119.00	2
9	92.08	0	113.20	1.25
10	89.61	0	111.80	1.25
11	89.61	0	111.80	1.25
12	89.07	0	110.78	1.25
13	88.62	0	110.27	1.25

It can be seen how, following OPS integration also in the existing terminal, max loading percentages rise, bringing the number of critical lines from six to eleven, and total overloading time-steps (over 100%) from 30 to 124. These results demonstrate of OPS integration is critical, and that if both ferry terminals are expected to accommodate vessels, the deployment of PV and BESS also in the existing terminal will be essential to mitigate network congestion.

C.3 Comparison of the PV and BESS Cost-Minimization Strategy

This section serves as a brief comparison between the new Ferry Terminal's scenarios and optimization strategies.

Particularly, Figures 41 and 42 show the difference in maximum line loading percentages by scenario, together with their maximum consecutive hours over 100%.

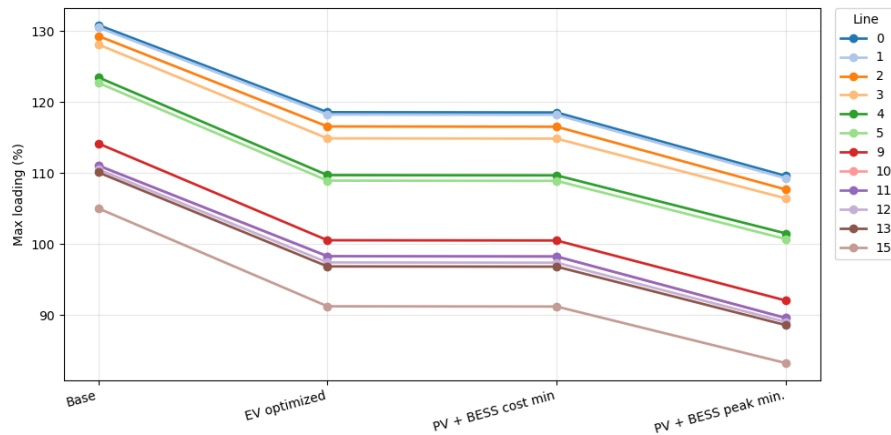


Figure 41: Maximum loading percentages by scenario

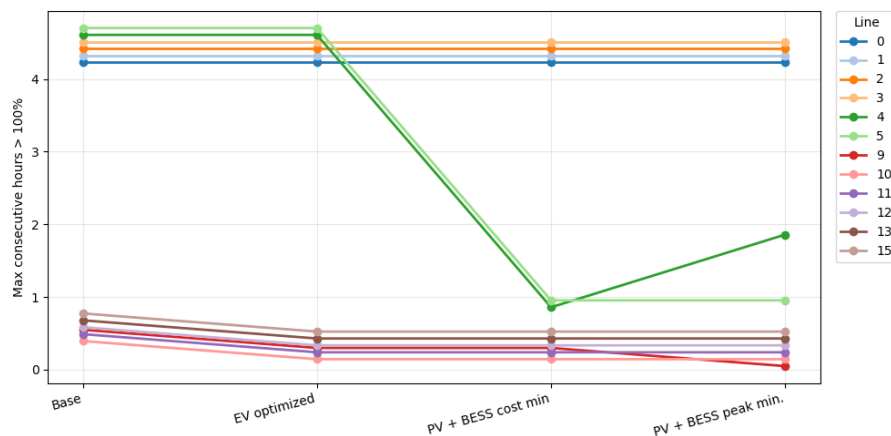


Figure 42: Maximum consecutive hours over 100% by scenario

As can be evaluated by also looking at Table C.6, the peak minimization scenario has the best performance in terms of grid impact mitigation, and it is therefore utilized for this report's simulations.

Table C.6: Total overloaded time steps for the different electrification scenarios

Total overloading timesteps	Base	EV optimized	PV+BESS cost min.	PV+BESS peak min.
> 100%	158	90	78	30
> 80%	1796	1601	1293	858